SCHOOL OF BUSINESS AND MANAGEMENT SCIENCES
DEPARTMENT OF FINANCIAL ENGINEERING

A HYBRID FRAMEWORK FOR PRICING EUROPEAN OPTIONS UNDER THE HESTON MODEL USING THE BATES METHOD

BY

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THIS RESEARCH IS SUBMITTED TO THE DEPARTMENT OF FINANCIAL ENGINEERING SCHOOL OF BUSINESS AND MANAGEMENT SCIENCES IN PARTIAL FULFILMENT OF THE BACHELOR OF TECHNOLOGY HONOURS DEGREE IN FINANCIAL ENGINEERING

2025

Title Page

Title of project:	A HYBRID FRAMEWORK FOR PRICING EUROPEAN OPTIONS UNDER THE HESTON MODEL USING THE BATES METHOD.
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Year:	2025

Declaration

I the author, Takudzwa Oscar Nyakudya, do hereby declare that this work has not previously been accepted in substance for any degree and is not being concurrently submitted in candidature for any degree.

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Dedication

This research project is dedicated to my mother, Mrs. Nyakudya, and my father, Mr. Nyakudya, whose unwavering support has been instrumental in my education. I also extend this dedication to my family and friends, whose encouragement and inspiration continue to guide me.

Acknowledgements

I would like to express my heartfelt gratitude to everyone who contributed to the successful completion of this dissertation. First and foremost, I extend my deepest thanks to the Almighty God for His boundless grace, kindness, and love. His guidance and strength have been my pillar throughout this journey. I am also profoundly grateful for the wonderful friends He has placed in my life. Their unwavering support, encouragement, and companionship have been invaluable, making this journey both fulfilling and memorable.

Secondly, I sincerely thank my supervisor, Mrs. P. Mawire, for her invaluable guidance and support throughout this project. Her insightful feedback and encouragement were essential to the completion of this dissertation.

Furthermore, I am deeply grateful to Vimbainashe Chikwava, Kudah Machiwana and Tafadzwa Maseka for their support and insightful contributions from the project's inception to its completion. Their guidance and encouragement have been instrumental in helping me overcome challenges and successfully navigate the research process.

I would also like to express my heartfelt gratitude to Mr. B. Handireketi, my high school teacher, for his consistent support and encouragement. Throughout this journey, he remained in touch, checking in on my progress and offering valuable advice. His belief in me has been a constant source of motivation, and I am truly thankful for his guidance and care.

Finally, I am deeply grateful to my family, especially my mother and my father, for their unwavering support and encouragement. Their belief in me kept me focused and motivated throughout this journey.

Thank you all for your unwavering help and support throughout this journey. Your encouragement, guidance, and assistance have been invaluable in helping me reach this milestone. I am deeply grateful for each of you and for the role you played in making this achievement possible.

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Abstract

This study designed a numerical framework for pricing European options by integrating the Bates jump-diffusion model with the Heston stochastic volatility model. The model's accuracy was evaluated using metrics such as Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), all of which demonstrated a close approximation to market price distributions. The framework's practical applicability was validated using real financial data, and its accuracy was compared to the Black-Scholes. Additionally, a comprehensive sensitivity analysis was conducted to examine the effects of changes in underlying asset prices and other key parameters on option pricing. Furthermore, a hedging strategy incorporating delta hedging and volatility adjustments was implemented to assess risk management effectiveness within the proposed framework. This research contributes significantly to financial engineering by enhancing the understanding of option pricing under stochastic volatility and jump-diffusion dynamics while providing insights into effective hedging techniques for mitigating risk.

KEY WORDS: Heston model, Bates model, stochastic volatility, jump-diffusion, European options, hedging, option pricing.

Abbreviations

BSM	Black-Scholes-Merton
CBOE	Chicago Board Options Exchange
FFT	Fast Fourier Transform
FINSEC	Financial Securities Exchange
GBM	Geometric Brownian Motion
MAPE	Mean Absolute Percentage Error
PDE	Partial Differential Equation
PIDE	Partial Integro-Differential Equation
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
SDE	Stochastic Differential Equation

Chapter 1

INTRODUCTION

Financial engineering is a discipline that applies mathematical and computational techniques to solve problems in finance, particularly in the development of models that aim to minimize risk and optimize returns (Neely, 2003). This field encompasses the creation of various financial models including pricing models, forecast models and risk management models which are essential for accurately valuing financial assets, including derivatives like options. According to (Hull, 2018), options are financial derivatives that grants the holder the right but not the obligation to either purchase or sell an underlying asset at a predetermined price within a specified time frame. This optionality makes them distinct from many other financial instruments. Over time various mathematical models have been proposed to reflect the complex nature of financial markets and to establish reliable option valuation frameworks. Classical models like Black-Scholes assume continuously evolving asset prices and often fail to incorporate sudden market movements such as those triggered by unexpected

economic developments, breaking news, or drastic changes in investor sentiment (Hull, 2018). To address these limitations, the Heston model was introduced a stochastic volatility framework that models the variability of volatility itself offering a more robust structure for pricing options under realistic market conditions (Heston, 1993).

The Heston model improves option pricing by introducing stochastic volatility capturing both price movements and volatility shifts. While this makes the model more realistic, it also complicates numerical implementation as methods like finite difference and finite element often face difficulties handling the stochastic components (Düring and Pitkin, 2017).

In order to overcome the difficulties presented by stochastic volatility, this study uses the Bates method for pricing European options within the Heston model framework. Known for its effectiveness in handling sharp variations in fields like fluid dynamics, the Bates method is well suited for capturing complex market behavior (Hull, 2017). It enhances the Heston model by including price jumps reflecting abrupt market shifts.

1.1 Background to the Study

1.1.1 History of Options

The origins of options can be traced back to ancient civilizations such as Greece and Rome, where contracts resembling modern options were

used to lock in future transactions (Chance and Brooks, 2015). A notable example involves the Greek philosopher Thales, who secured rights to olive presses ahead of the harvest season an early demonstration of a call option strategy. In the 17th century, more structured forms of options trading appeared in Amsterdam, notably tied to tulip bulb speculation during the period known as Tulip Mania. However, these markets were largely unregulated and highly susceptible to speculation and price bubbles (Chance, 2008).

The launch of the Chicago Board Options Exchange (CBOE) on April 26, 1973, marked the beginning of regulated and standardized options trading, transitioning the market away from informal over-the-counter transactions (Chance, 2008). This advancement was accompanied by the BlackScholes-Merton model's debut, which offered a ground-breaking framework for European-style option pricing. Together, these events catalyzed the growth and sophistication of global derivatives markets.

1.1.2 Introduction to Options

An option is a financial instrument that provides the holder with the right without any obligation to purchase or sell an asset at a designated price during a specified time frame according to (Hull, 2018). An option's underlying asset may range from financial instruments like stocks, bonds, and indices to commodities.

Premium

According to (Hull, 2017) premium is the upfront, non-refundable fee paid

by the buyer for the rights granted by the option. If exercised, the seller must honor the contract terms at the predetermined exercise price. For the buyer, the premium represents the cost of flexibility while for the seller, it's compensation for the potential obligation.

Strike Price:

Predetermined value to purchase the referenced base asset.

Expiration Date:

The date after which the option becomes worthless if not exercised.

Payoff diagrams:

Payoff diagrams facilitate the observation of how an option's value responds to changes in the price of the base asset. They also illustrate the payoff structure of complex strategies involving multiple positions. Analyzing these diagrams provides practical insights into the value of multi-leg strategies. Intrinsic value only applies to in-the-money options, and it matches their market price at expiration.

The payoff pattern for call and put options with a $25 exercise price is displayed in Figure 1.1 above. Any movement in the primary security price above $25 leads to a proportionate increase in the call option's intrinsic value, represented by a solid black line. The call loses all of its value if the price of base asset drops below $25. The intrinsic value of the put option is shown by the dashed gray line and it rises when the reference instrument price falls below $25. The put option fully loses value above $25.

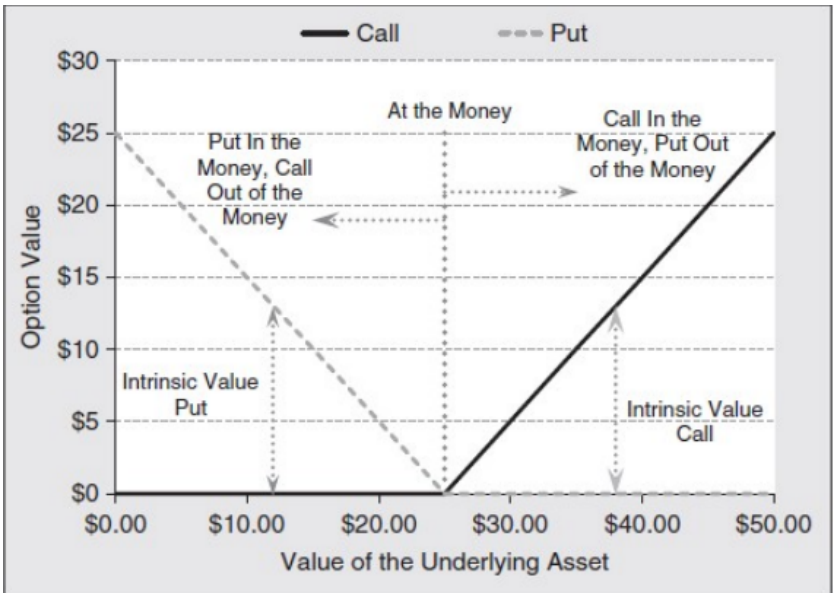

Figure 1.1: Pay Off Diagram

1.1.3 Uses and types of Options

Options are used in portfolio management and investment strategies for several purposes:

- **Hedging:** Options protect against losses.
- **Income Generation:** In the event that owned equities remain below the strike price, selling covered calls on them yields premiums.
- **Speculation:** Traders use options to bet on price movements, profiting from rises with calls and declines with puts. Options also provide leverage for larger positions with less capital.

The two main categories of financial options are called calls and puts and they are selected according to investment objectives and risk management.

- **Call Options:** Bullish instruments allowing the buyer to purchase at a set price. For example, with a $25 strike price and $1 premium, the

option is exercised if the stock exceeds $26, allowing the investor to profit by buying at $25 and selling at a higher price

- **Put Options:** Bearish tools that grant the buyer the authority to sell at a predetermined price (Hull, 2018). For example, with a $25 strike price and $1 premium the option is exercised if the stock falls below $24 allowing the investor to sell at $25 and buy at a lower market price.

1.1.4 Models used to price Options

The Black-Scholes model, introduced in 1973, was the first major framework for pricing European options. The model assumes constant volatility, no dividends, and continuous trading. Despite being widely utilized, its shortcomings have led to the creation of more sophisticated models.

Limitations of the Black-Scholes Model:

- Fixed volatility, which does not represent actual market conditions.
- No Dividends: The model doesn't account for dividends, which are common in real-world financial markets.
- Ignores dividends paid by underlying assets.
- Designed only for European options (not suitable for American options).
- Assumes a fixed, constant risk-free interest rate.
- Excludes transaction costs, taxes and other market frictions.

Alternative Models to Address Black-Scholes Limitations:

1. **Black-Scholes-Merton Model:** Incorporates a dividend yield for pricing options on stocks that pay dividends.
2. **Heston Model:** Accounts for changing market volatility by making it a random process that fluctuates over time.
3. **Bates Model:** In contrast to the Heston model, which only addresses changing volatility, the Bates model includes random jumps in price, reflecting more realistic market behavior.
4. **Binomial Model:** Uses a price tree to model discrete movements suitable for pricing both European and American options.
5. **SABR Model:** Captures volatility skew; useful for pricing interest rate derivatives with dynamic volatility.
6. **Jump-Diffusion Models:** Combines continuous diffusion with sudden jumps in price behavior.

1.1.5 Introduction of Options in Zimbabwe

In 2022, Zimbabwe's FINSEC introduced European style options. The Automated Trading Platform supports this market involving contract writers, clearing participants, stockbrokers and custodian banks. These roles ensure smooth trading, collateral security, and market integrity, contributing to the growth of Zimbabwe's capital markets and investor confidence.

Currently, FINSEC offers only European style stock options exercisable exclusively at expiration. These options are tailored for investors seeking

strategies tied to specific maturity dates. The details of these transactions are summarized in the Figure 1.2 and 1.3 below.

Category of Contract	Stock Options
Underlying Financial Instrument	Delta Corporation, Innscor Limited, Simbisa Brands, Econet Wireless, Ecocash Holdings
Expiry Dates	Determined by the option writer but last Thursday of each month
Contract Months	Determined by the option writer
Expiry Times	12.00 CAT
Listing Program	Quarterly
Valuation Method on Expiry	Based on the volume weighted average price (VWAP) of the underlying instrument for liquid contracts and theoretical price (spot + cost of carry) for illiquid contracts
Methodology for Settlement	Cash Settlement
Contract Size	100 shares per contract

Figure 1.2: Details of Stock Option Contracts (Source: finsec.co.zw)

Figure 1.2 above provides comprehensive details on stock option contracts, including underlying assets, expiration, and cash-settled valuation. Figure 1.3 below lists the corresponding market fees (levies, brokerage, VAT), which are calculated on the notional value. Both figures are essential for understanding contract components and trading costs.

Charge Types	Buying and Selling fees (%)
Sec Levy	0.05%
IPF Levy	0.025%
Platform Levy	0.14%
Guarantee Fund Levy	0.025%
Brokerage Fees	0.14%
VAT @15% of Brokerage Fees	0.0145%
The percentages above are used to calculate fees based on the notional contract value.	

Figure 1.3: Market Fees for Stock Options (Source: finsec.co.zw)

Overview of Trading Volume and Value in Zimbabwe

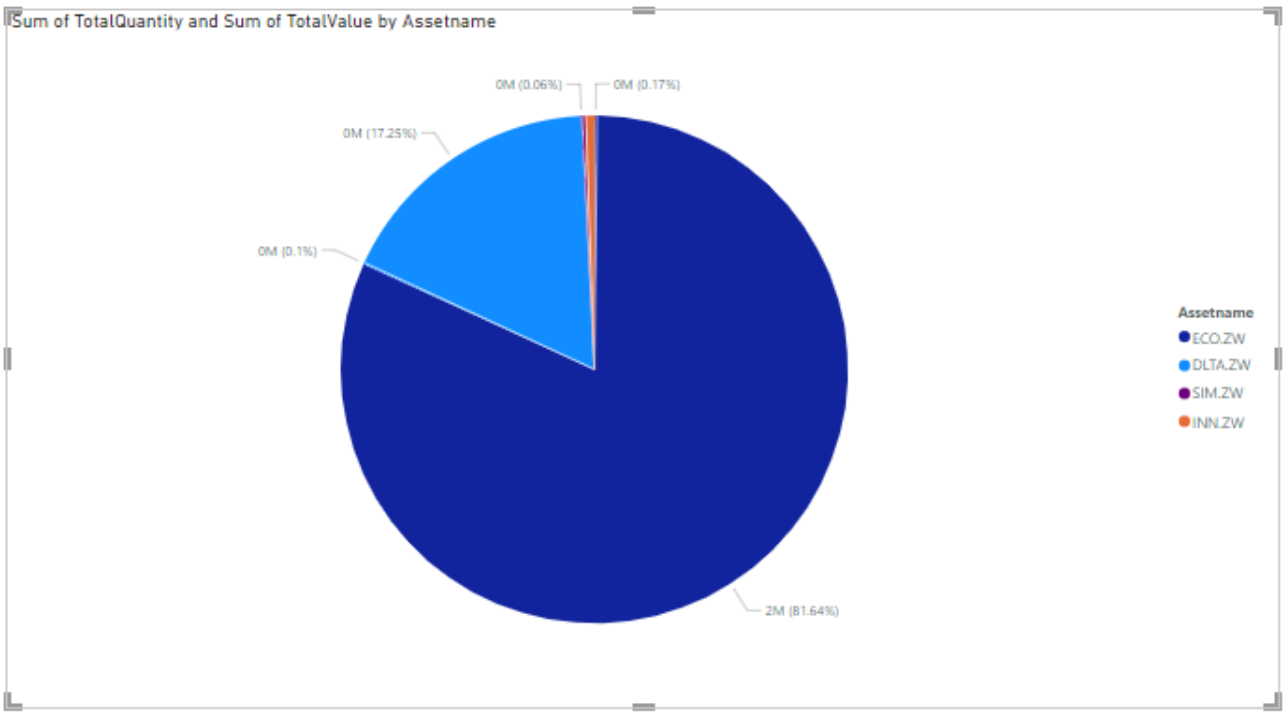

Figure 1.4: Trading Volume and Value of Stock Options in Zimbabwe (Source: finsec.co.zw)

1.2 Statement of the Problem

Accurate pricing of European options is challenged by increasing market volatility and complexity. Traditional models like Black-Scholes neglect critical risks from stochastic volatility and price jumps, a limitation starkly revealed during financial crises. While improved models address these factors individually, a hybrid approach combining them remains under-explored.

This is particularly relevant in Zimbabwe's emerging derivatives market, where economic policies cause significant fluctuations. Reliance on the

Black-Scholes model has led to persistent mispricing. This study aims to bridge this gap by developing a hybrid framework integrating the Heston and Bates models to enhance pricing accuracy, market efficiency and investor confidence.

1.3 Research Objectives

This study seeks to address the above-stated problem with the following objectives.

1. To model a hybrid pricing framework for European options that integrates the Bates jump-diffusion process with the Heston stochastic volatility model.
2. To evaluate model accuracy in pricing European options compared to traditional models.
3. To assess model adaptability in varying volatility regimes and price jumps.
4. To investigate hedging techniques for option pricing using Bates-Heston model.

1.4 Research Questions

1. How can the Bates jump-diffusion process be integrated with the Heston stochastic volatility model to develop a unified pricing frame-

work for European options?

2. How does the Bates-Heston model perform in pricing European options compared to traditional models?

3. To what extent does the incorporation of both stochastic volatility and jumps in the Bates-Heston model enhance its adaptability in varying volatility regimes and price jumps?

4. What are the most effective hedging techniques for option pricing using the Bates-Heston model?

1.5 Research Hypothesis

H_0: Bates-Heston is accurate in valuing European options.

H_1: Bates-Heston is not accurate in valuing European options.

1.6 Significance of the Study

Financial practitioners such as portfolio managers and option traders frequently encounter mispricing risks that can lead to suboptimal investment decisions. This study introduces a hybrid framework that find option value that incorporates both random volatility fluctuations and jump diffusion dynamics features commonly observed in real financial markets to enhance pricing precision. By evaluating the model's performance against established frameworks such as Black-Scholes and the standard Heston model the research provides practical guidance for selecting appropriate models under varying market conditions. Traditional models often fall

short in capturing complex market phenomena including volatility clustering, abrupt price movements and heavy tailed return distributions. The proposed approach aims to address these limitations offering a more robust and realistic framework for accurately valuing European-style options in volatile market environments.

1.7 Scope of the Study

1. **Model Integration:** Combines Bates jump-diffusion and Heston stochastic volatility to improve European option pricing.
2. **Numerical Methods and PDEs:** Utilizes FFT and Monte Carlo simulations to solve PDEs with a focus on accuracy and efficiency.
3. **Market Conditions:** Assesses model performance across different volatility and jump scenarios.
4. **Comparative Analysis:** Compares Bates-Heston accuracy with traditional models.
5. **Empirical Validation:** Validates model using real market data to ensure reliability.
6. **Practical Implications:** Explores use in trading, investment and risk management

1.8 Assumptions of the Study

i. **Efficient Markets:** Assumes asset prices fully reflect all relevant information, enabling accurate pricing with the Bates-Heston model.

ii. **Constant Parameters:** Stochastic volatility and jump-diffusion parameters remain fixed during the pricing period.

iii. **Asset Dynamics:** Asset prices follow a log-normal distribution, with jumps being independent and identically distributed.

iv. **Risk-Neutral Valuation:** Pricing is carried out under risk neutral measure where returns are equivalent to the risk free rate.

v. **No Market Frictions:** Assumes no transaction costs or taxes, focusing purely on theoretical pricing.

1.9 Limitations of the Study

i. **Simplifying Assumptions:** Assumes constant parameters and no transaction costs, limiting real-world applicability.

ii. **Parameter Calibration:** Pricing accuracy depends heavily on precise calibration of model parameters.

iii. **Data Dependence:** Findings rely on the availability and quality of empirical market data.

iv. **Computational Complexity:** The model's integration requires complex numerical methods, increasing computational cost.

v. **Scope Limitation:** Focuses solely on European options, excluding

American and exotic options.

1.10 Organization of the Study

The research is structured into five chapters:

- **Chapter One** introduces the study, covering the objectives, background, hypothesis, problem statement, research questions, significance, scope and limitations.
- **Chapter Two** presents a review of related literature, highlighting empirical findings, theoretical models and research gaps.
- **Chapter Three** outlines the research methodology including data collection techniques and analytical models.
- **Chapter Four** delves into the analysis of data and the interpretation of the model's results.
- **Chapter Five** discusses the findings, provides recommendations and conclusion of the research.

1.11 Conclusion

This chapter outlines the study's aim, objectives, significance for stakeholders, limitations and assumptions that define the research conditions.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

In this chapter, we introduce the conceptual framework and discuss relevant theoretical and empirical studies on pricing options of European options under the hybrid framework integrating the Heston and Bates models. It focuses on theoretical contributions discussing various numerical methods applied to these models for option pricing. Additionally, the chapter examines empirical studies by other researchers, highlighting their findings and the effectiveness of incorporating stochastic volatility and jump-diffusion dynamics in option pricing.

2.2 The Conceptual Framework

A conceptual framework is a structure, which the researcher believes can best explain the natural progression of the phenomenon to be studied

(Camp, 2001).

Dependant variable

Option Price : As the dependent variable, the option price quantifies the market value of a European option obtained by solving the Bates model's governing partial differential equation using specified independent variables (Hull, 2018).

Independent variables

Time to Expiration (T): Denotes the interval remaining before option maturity.

Volatility(σ)
Indicates the fluctuation of gains on the referenced base asset, mirroring the sensitivity of price differences and perceived market risk.

Interest Rate (r) Reflects the discount rate, which affects future payouts' present value. Over the term of the option, it is presumed to remain constant.

Strike Price (K) Indicates the predetermined level at which the option owner can exercise their permission to buy or sell the backing asset.

Jump Intensity (λ)
Jump intensity (λ) illustrates the frequency of price jumps, which are depicted using a Poisson distribution to highlight unforeseen market fluctuations. Depending on the jumps occurring in accordance with a Poisson process of intensity (λ), the probability mass function of observing k jumps

over time period t is given by the probability mass function of:

$$P(N_t = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!} \tag{2.1}$$

Jump Size (η)

Denotes the magnitude of discrete asset price movements due to jumps and is typically described using a double exponential distribution, enabling asymmetric modeling of positive and negative jumps. The associated probability density function (PDF) is:

$$f(\eta) = \frac{\lambda}{2} e^{-\lambda|\eta-\mu|} \tag{2.2}$$

where:

- μ is the mean of the jump sizes.

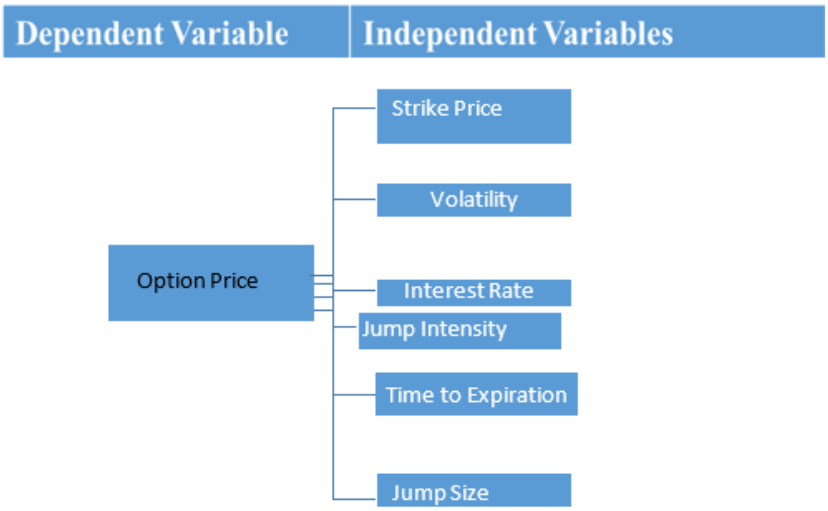

Figure 2.1: How variables connect to the option price

2.3 Theoretical Literature

European option valuation in this study is conducted through the Bates-Heston framework, which merges stochastic volatility with jump-diffusion processes and relies on foundational concepts in financial mathematics.

2.3.1 Girsanov's Theorem: Measure Change

Girsanov's theorem provides a method for changing the probability measure from $\mathbb{P}$, the real-world measure, to $\mathbb{Q}$, the risk-neutral measure, by adjusting the drift component of the stochastic process ((Shreve, 2004)). For a Brownian motion $W_t^{\mathbb{P}}$ under measure $\mathbb{P}$, the corresponding process $W_t^{\mathbb{Q}}$ under measure $\mathbb{Q}$ satisfies the equation:

$$dW_t^{\mathbb{Q}} = dW_t^{\mathbb{P}} - \theta_t \, dt,$$

The Radon-Nikodym derivative that ensures this transformation is expressed as:

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \mathcal{E}\left(-\int_0^t \theta_s \, dW_s^{\mathbb{P}}\right),$$

where $\mathcal{E}(\cdot)$ denotes the stochastic exponential. The stochastic exponential of a continuous semimartingale X_t is defined as:

$$\mathcal{E}(X_t) = \exp\left(X_t - \frac{1}{2}\langle X\rangle_t\right), \tag{2.3}$$

with $\langle X\rangle_t$ representing the quadratic variation of the process X_t (Shreve, 2004).

Application to Heston Model

The Heston model describes the stock price S_t and variance v_t under $\mathbb{P}$ as:

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \tag{2.4}$$

$$dv_t = \kappa(\theta - v_t)dt + \sigma_v \sqrt{v_t} dW_t^{(2)}, \tag{2.5}$$

Under the risk-neutral measure $\mathbb{Q}$, Girsanov's theorem gives:

$$dW_t^{(1),\mathbb{Q}} = dW_t^{(1),\mathbb{P}} - \frac{\mu - r}{\sqrt{v_t}} dt, \tag{2.6}$$

$$dW_t^{(2),\mathbb{Q}} = dW_t^{(2),\mathbb{P}} - \lambda_v dt. \tag{2.7}$$

Thus, the risk-neutral dynamics become:

$$dS_t = r S_t dt + \sqrt{v_t} S_t dW_t^{(1),\mathbb{Q}}, \tag{2.8}$$

$$dv_t = \kappa^*(\theta^* - v_t)dt + \sigma_v \sqrt{v_t} dW_t^{(2),\mathbb{Q}}. \tag{2.9}$$

Application to Bates Mode

In a jump-diffusion model, such as the Bates model, the stock price follows:

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)} + J_t S_t dN_t, \tag{2.10}$$

where N_t follows a Poisson process governed by intensity λ, and $J_t \sim e^Z - 1$ with $Z \sim \mathcal{N}(\mu_J, \sigma_J^2)$. Applying Girsanov's theorem, the risk-neutral dynamics adjust to:

$$dS_t = (r - q - \lambda \mathbb{E}^{\mathbb{Q}}[J])S_t dt + \sqrt{v_t} S_t dW_t^{(1),\mathbb{Q}} + J_t S_t dN_t^{\mathbb{Q}}, \tag{2.11}$$

where the jump intensity transforms as:

$$\lambda^{\mathbb{Q}} = \lambda e^{-\nu}, \tag{2.12}$$

with ν as the risk premium for jumps.

2.3.2 Poisson Process of Jumps

Jumps in the financial markets are modeled by a Poisson process N_t. It is defined by:

$$P(N_t = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k = 0, 1, 2, \ldots, \tag{2.13}$$

The key properties of a Poisson process include:

- $N_0 = 0$ almost surely.
- Independent increments.
- Stationary increments.

The probability of a single jump happening within a small time frame Δt is approximately:

$$P(N_{t+\Delta t} - N_t = 1) \approx \lambda \Delta t, \tag{2.14}$$

while the probability of more than one jump is negligible:

$$P(N_{t+\Delta t} - N_t > 1) \approx 0. \tag{2.15}$$

A compound Poisson process is employed to model jumps in asset prices, with the total magnitude of a jump represented by:

$$X_t = \sum_{i=1}^{N_t} J_i, \tag{2.16}$$

where J_i are jump sizes that are independently distributed and have the same distribution. The Poisson-driven jump process is widely used in financial models like the Merton and Bates models to represent sudden market movements.

When pricing under the risk-adjusted probability measure $\mathbb{Q}$, jump intensity and distribution are typically modified to reflect risk premiums. The transformed jump intensity is typically given by:

$$\lambda^{\mathbb{Q}} = \lambda e^{-\nu}, \tag{2.17}$$

where ν is the risk premium associated with jumps.

2.3.3 Brownian Motion

A crucial stochastic process in finance, Brownian motion, represented by W_t, is frequently utilized to describe asset dynamics and option pricing. It evolves continuously over time and possesses distinct stochastic characteristics described as follows:

- $W_0 = 0$ almost surely.
- Independent increments: Increments $W_{t_{i+1}} - W_{t_i}$ over non-overlapping intervals are independent.
- Normally distributed increments.
- The process W_t evolves continuously over time with probability one.

Brownian motion is central to stochastic calculus and is used to model random movements in financial markets. A geometric Brownian motion is commonly used to describe market prices of shares:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \tag{2.18}$$

where:

- S_t - asset price
- μ - drift
- σ - volatility
- W_t - standard Brownian motion.

This SDE forms the basis of the BSM for option pricing.

Brownian Motion Under Measure Change

Girsanov's theorem is used to transform Brownian motion under a risk-adjusted probability measure $\mathbb{Q}$. Under the real-world measure $\mathbb{P}$, if Wt is a Brownian motion, then under $\mathbb{Q}$:

$$dW_t^{\mathbb{Q}} = dW_t^{\mathbb{P}} - \theta dt, \tag{2.19}$$

where the market price of risk is represented by θ. In order to price derivatives without arbitrage, this transformation makes sure that discounted asset values become martingales under $\mathbb{Q}$.

Theorem 2.1 Assuming that asset prices move in a geometric Brownian motion, the Black-Scholes-Merton model is essential to option pricing. However, it does not account for jumps, limiting its ability to capture market shocks. While foundational the BSM model is extended by stochastic volatility and jump-diffusion models like Bates-Heston. The asset price dynamics are given by:

$$dS(t) = \mu S(t)dt + \sigma S(t)dW(t) \tag{2.20}$$

The foundation of European option valuation, according to the Black-Scholes-Merton (BSM) framework, is the Black-Scholes equation. The model's assumptions, including constant volatility, have led to the creation of expanded models that include stochastic volatility and jump dynamics, despite its broad use. According to the BSM model, a European call option's

pricing is typically calculated as follows:

$$C = S(t)N(d_1) - Ke^{-r(T-t)}N(d_2)$$

where $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution (Hull, 2017).

Theorem 2.2 The Bates-Heston model expand the Black-Scholes-Merton (BSM) framework by incorporating stochastic volatility and jump-diffusion processes, enabling it to capture both volatility clustering and discontinuous price changes (Bates, 1996). The price of an asset is governed by a jump-diffusion process incorporating a compound Poisson component, where the frequency and magnitude of jumps are controlled by defined parameters.

The Bates-Heston framework is characterized by a set of stochastic differential equations that define the model's behavior:

$$dS(t) = (\mu - \lambda J - 0.5\sigma^2)S(t)dt + \sigma S(t)dW(t) + JdN(t) \tag{2.21}$$

where:

- $dS(t)$ = shifts in price of asset $S(t)$
- μ = expected return
- λ represents the frequency at which jumps occur in the process
- J denotes the magnitude of each jump in the process
- σ = volatility,

- dt = infinitesimal time increment
- $dW(t)$ = standard Wiener process
- $JdN(t)$ = jump component, where $dN(t)$ follows a Poisson process.

2.3.4 Common Numerical Methods for Option Valuation

Theorem 2.3 The Monte Carlo method generates random asset price trajectories computes the corresponding option payoffs for each path and then averages the results to approximate the option's value.

1. **Monte Carlo simulation equation:**

$$S_{ij} = S_{i,j-1} \cdot e^{(r-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}Z_{ij}} \tag{2.22}$$

where:

- S_{ij} denotes the value of the asset at the j-th time step in the i-th simulation path.
- $S_{i,j-1}$ is the asset price at the previous time step,
- r is the safe rate of return
- σ indicates the asset's volatility
- Δt is time step length
- Z_{ij} is a standard normal random variable.

2. **Option payoff calculation:**

$$V_i = \max(S_{ij} - K, 0) \tag{2.23}$$

where:

- V_i refers to the option payoff in the i-th simulation,
- K represents the strike price of the option.

3. **Option value estimation:**

$$V = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^{N} V_i \tag{2.24}$$

where:

- V stands for the option's expected value
- T denotes the option's time until expiration
- N represents the total number of simulations.

Monte Carlo simulations are especially useful for pricing different types of options such as exotic and path dependent options. They can also accommodate models with multiple underlying assets and stochastic volatility. However, achieving higher accuracy in these simulations demands substantial computational power and challenges with convergence may occur when dealing with infrequent events.

Theorem 2.4 The Binomial tree technique represents asset price movements using a stepwise framework involving two possible directions up

or down. This tree is traversed from the last time step back to the first time step in order to compute option values recursively.

1. **Binomial tree model equation**: The asset price at any given node (i, j) in the tree is calculated as:

$$S_{ij} = S_0 \cdot u^i \cdot d^{j-i} \tag{2.25}$$

Where:

- $S_{i,j}$ denotes the asset price at node (i, j), where i is the number of upward moves and j is the total time steps
- S_0 is the initial price of the underlying asset
- u is the price increase factor in an up move.
- d is the factor by which the price decreases in a down move
- i indicates the count of up moves.
- j denotes the total time steps

2. **Risk-neutral probability**:

$$q = \frac{e^{r \cdot \Delta t} - d}{u - d} \tag{2.26}$$

Where:

- q represents the upward price probability.
- r denotes the continuously compounded safe return.
- Δt indicates the time interval between steps in the model.

3. **Option valuation recursion:**

The value of the option at any given node (i, j) is determined by the following recursion:

$$V_{ij} = e^{-r \cdot \Delta t} \cdot (q \cdot V_{i+1,j+1} + (1 - q) \cdot V_{i,j+1}) \tag{2.27}$$

Where:

- V_{ij} represents the calculated option price at the node corresponding to the i-th upward movement and the j-th time step.

Binomial tree method is a simple and intuitive approach, suitable for various option types, including American options. However, it may require high precision and many time steps, leading to computationally intensive results (Ahmed Kamel and Elkarmoty., 2023).

Theorem 2.5 Characteristic Function of the Bates Model and Option Pricing

Bates model is a development of the Heston model that adds jump-diffusion processes and stochastic volatility. European options are priced using the characteristic function $\varphi(u)$ of the Bates model, which has terms that take into consideration both the jumps and stochastic volatility.

$$\varphi(u) = \mathbb{E}\left[e^{iu \log(S_T)}\right]$$

Where:

- u represents the value of a European option with a specified exercise

price.

- $\log(S_T)$ is the logarithm of the asset price at maturity T.

The $\varphi(u)$ is derived by solving the corresponding partial differential equations (PDEs) of the Bates model. Once the characteristic function is available, European option prices can be computed using Fourier transform techniques.

A popular method to compute the option price in the Fourier domain is the **Carr-Madan method**. The value $C(K)$ of a European option with exercise price K is determined by the by the inverse Fourier transform of the characteristic function:

$$C(K) = e^{-rT} \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{iu\log(K)}\varphi(u-i)}{u^2+1} du$$

Where:

- $\varphi(u)$ is the characteristic function of the Bates model.

This approach allows for an efficient calculation of the option price, even for complex models such as the Bates model, and is particularly useful when dealing with jump-diffusion models.

Theorem 2.6 Closed-Form Solution of the Bates Model PDE

The Bates model incorporates both stochastic volatility and jump-diffusion components, governed by a system of PDEs (Bates, 1996).An analytical solution for the option price can be obtained by solving these PDEs using Fourier transform techniques, specifically through the characteristic func-

tion $\varphi(u)$.

The PDE governing the Bates model is generally of the form:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0$$

Where:

- C = option price,
- S = asset's value,
- r = risk-free yield.
- σ = volatility,
- t = time.

The solution to this PDE, in the case of a European call option, can be efficiently obtained by leveraging the characteristic function $\varphi(u)$, which is derived by solving the corresponding system of equations for the Bates model.

For European call options, the closed-form solution of the PDE is given by the inverse transform of the characteristic function $\varphi(u)$:

$$C(K) = e^{-rT}\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{e^{iu\log(K)}\varphi(u-i)}{u^2+1}du$$

Where:

- $\varphi(u)$ is the characteristic function of the Bates model.

This solution allows for the efficient computation of the option price without requiring direct numerical solution of the PDE, making it a valuable tool for pricing options under the Bates framework.

Theorem 2.7 Fast Fourier Transform for Option Pricing

The Fast Fourier Transform (FFT) efficiently computes option prices by applying the inverse Fourier transform to the characteristic function $\varphi(u)$ of the asset price dynamics (Carr and Madan, 1999). For a European option, the price at strike K is given by:

$$C(K) = e^{-rT} \frac{1}{2\pi} \sum_{n=0}^{N-1} e^{iu_n \log(K)} \varphi(u_n - i) \frac{1}{u_n^2 + 1}$$

Where:

- $C(K)$ is the call option price,
- r is the risk-free rate,
- T is the time to maturity,
- u_n are the frequency points.

Using FFT, the computation complexity is reduced to $O(N \log N)$, making it efficient for large-scale pricing, especially in models like Bates.

2.3.5 Types of Options

European Option

According to (Hull, 2018), a European option provides the holder the right but not the obligation to execute a buy or sell transaction of the underly-

ing asset at a predefined strike price strictly at the option's expiration date. European options are financial derivatives that grant the holder the right but not the responsibility, to buy or sell an underlying asset at a fixed price on a given date, according to (Xiang, 2024). This contrasts with American options which can be exercised any time before expiration as noted by (Jamshidian., 1992).

American Option

An American option grants the holder the right to buy or sell an underlying asset at a specified price at any point up to and including the expiration date and provide the flexibility of early exercise (Hull, 2018). This feature makes them valuable tools for both dynamic hedging and speculative strategies. They are traded on exchanges as well as over-the-counter markets, and their pricing is influenced by various factors such as the underlying asset's price, volatility, time remaining to expiration, and prevailing interest rates (Hull, 2017).

Exotic Options

They include barrier options, which activate or expire when the underlying price reaches certain levels, and Asian options, which depend on the average price over time (Hull, 2018). Other types, like digital and chooser options, offer fixed payouts or allow the holder to choose the option type later. These options are commonly used for advanced hedging or speculative strategies and are traded over-the-counter, requiring more complex

valuation and risk management than standard options (Hull, 2017).

2.3.6 Classification of Options

Call Option

A call option gives the holder the right to buy the underlying asset at the strike price on or before the expiration date according to (Hull, 2017).

Put Option

A put option gives the holder the right to sell the underlying asset at the strike price on or before the expiration date according to (Poitras., 2009).

2.3.7 Option Payoff Functions:

European Call Option:

The payoff is:

$$C = \max(S_T - K, 0)$$

Where:

- C = payoff of the call option,
- S_T = asset price at expiration,
- K = strike price.

European Put Option:

The payoff is:

$$P = \max(K - S_T, 0)$$

Where:

- P = payoff of the put option,
- S_T = asset price at expiration,
- K = strike price.

2.3.8 Hedging

According to (Ross., 1998), hedging is a risk management strategy used to reduce or eliminate losses from unfavorable price fluctuations. In option pricing and financial derivatives, hedging involves creating offset positions to minimize or eliminate exposure to price risk, aiming to balance the portfolio against unfavorable price movements (Chen-Miao Lin and Smith., 2008).

The method of applying hedging in the Bates model for option pricing involves constructing a portfolio of assets to offset the risk associated with the option position (Ross., 1998). According to (Düring and Pitkin, 2017), the core concept of hedging in this context is to build a portfolio, often composed of the underlying asset and risk-free assets, such that their combined value moves in a way that reduces the risk associated with the option position.

Delta-Vega hedging adjusts the portfolio to keep both delta and vega near zero, reducing risk from price fluctuations and volatility changes. The required number of shares for delta hedging is governed by the option's delta. and the number of options held, while an additional option position is introduced to hedge vega risk. According to (**?**), delta fluctuates with changes in the underlying asset price, and vega is affected by shifts in implied volatility.

2.3.9 Traditional Option Pricing Models

Black-Scholes-Merton Pricing Model

The Black-Scholes model, introduced in 1973, offers a method for pricing options based on assumptions of constant risk-free interest rates, log-normal asset price distribution, and the absence of market frictions (Black and Scholes, 1973*a*). However, its assumptions particularly those of constant volatility and market efficiency may not always hold in real market conditions. The Black-Scholes equation describes the dynamics of asset prices under these idealized assumptions (Black and Scholes, 1973*b*):

$$dS(t) = \mu S(t)\,dt + \sigma S(t)\,dW(t)$$

where $S(t)$ is the asset price at time t, μ is the drift rate, σ is the volatility, and $dW(t)$ represents a Wiener process.

Assumptions of the Black-Scholes model:

1. Asset follows geometric Brownian motion.

2. Constant volatility.

3. Constant risk-free interest rate.

4. No transaction costs or taxes.

5. No jumps.

The **Black-Scholes formula** for option pricing is:

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2) \tag{2.28}$$

$$P = Ke^{-rT} N(-d_2) - S_0 N(-d_1) \tag{2.29}$$

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}} \tag{2.30}$$

$$d_2 = d_1 - \sigma\sqrt{T} \tag{2.31}$$

where:

- S_0 = Current stock price
- K = Strike price
- T = Time to maturity (in years)
- r = Risk-free interest rate
- σ = Volatility of the underlying asset

- $N(d)$ = CDF of the standard normal distribution

The formula's assumptions can limit its accuracy in dynamic markets, leading to the integration of models like Heston and Bates for more realistic pricing through stochastic volatility and jumps.

The Heston Model

Heston model, introduced in 1993, enhances the Black-Scholes framework by incorporating time-varying volatility. It models volatility as a stochastic process, allowing it to capture empirical features such as volatility clustering and skewness commonly observed in financial markets. In the Heston model, the asset price and its variance are governed by the following system of stochastic differential equations (Heston, 1993)

Asset Price SDE:

$$dS_t = \mu S_t \, dt + \sqrt{V_t} \, S_t \, dW_t^S$$

Variance SDE:

$$dV_t = \kappa(\theta - V_t) \, dt + \sigma_V \sqrt{V_t} \, dW_t^V$$

Heston Model PDE Derivation

The Heston model leads to the following partial differential equation for option pricing:

$$\frac{\partial P}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 P}{\partial S^2} + \rho \sigma v S \frac{\partial^2 P}{\partial v \partial S} + \frac{1}{2} \sigma^2 v \frac{\partial^2 P}{\partial v^2} + (r-q) S \frac{\partial P}{\partial S} + (a-bv) \frac{\partial P}{\partial v} - rP = 0. \tag{2.32}$$

Portfolio Construction

We construct a self-financing portfolio:

$$\Pi = f - \Delta S - hg. \tag{2.33}$$

Applying Itô's Lemma to f and g:

$$df = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S}dS + \frac{\partial f}{\partial v}dv + \frac{1}{2}vS^2\frac{\partial^2 f}{\partial S^2}dt + \frac{1}{2}\sigma^2 v\frac{\partial^2 f}{\partial v^2}dt + \sigma vS\rho\frac{\partial^2 f}{\partial S\partial v}dt, \tag{2.34}$$

$$dg = \frac{\partial g}{\partial t}dt + \frac{\partial g}{\partial S}dS + \frac{\partial g}{\partial v}dv + \frac{1}{2}vS^2\frac{\partial^2 g}{\partial S^2}dt + \frac{1}{2}\sigma^2 v\frac{\partial^2 g}{\partial v^2}dt + \sigma vS\rho\frac{\partial^2 g}{\partial S\partial v}dt. \tag{2.35}$$

The portfolio change is:

$$d\Pi = df - \Delta dS - hdg. \tag{2.36}$$

Risk-Neutralization:

$$\frac{\partial f}{\partial S} - h\frac{\partial g}{\partial S} - \Delta = 0, \quad \frac{\partial f}{\partial v} - h\frac{\partial g}{\partial v} = 0. \tag{2.37}$$

Setting $d\Pi = r\Pi dt$:

$$\left(\frac{\partial f}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 f}{\partial S^2} + \frac{1}{2}\sigma^2 v\frac{\partial^2 f}{\partial v^2} + \sigma vS\rho\frac{\partial^2 f}{\partial S\partial v}\right)dt - h\left(\frac{\partial g}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 g}{\partial S^2} + \frac{1}{2}\sigma^2 v\frac{\partial^2 g}{\partial v^2} + \sigma vS\rho\frac{\partial^2 g}{\partial S\partial v}\right) \tag{2.38}$$

Using Heston's risk-adjusted drift:

$$f(S, v, t) = -\gamma(\theta - v) + \lambda(S, v, t), \tag{2.39}$$

We obtain the Heston PDE:

$$\frac{\partial f}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 f}{\partial S^2} + \rho \sigma v S \frac{\partial^2 f}{\partial v \partial S} + \frac{1}{2} \sigma^2 v \frac{\partial^2 f}{\partial v^2} - r f + r S \frac{\partial f}{\partial S} + [\gamma(\theta - v) - \lambda(S, v, t)] \frac{\partial f}{\partial v} = 0. \tag{2.40}$$

The Bates Model

The Bates model enhances the Heston framework by incorporating jump components into the asset price process. While the Heston model captures stochastic volatility, it assumes continuous asset price changes. However, real financial markets often experience sudden jumps due to events. The Bates model addresses this limitation by introducing a jump component into the Heston framework. By integrating stochastic volatility with jumps, it offers a more accurate depiction of market behavior, particularly during periods of financial stress (Bates, 1996).

Bates Model Equations

Stock Price Dynamics:

$$dS_t = S_t \left(\mu \, dt + \sqrt{V_t} \, dW_t^S + (e^J - 1) \, dN_t \right)$$

Volatility Dynamics:

$$dV_t = \kappa(\theta - V_t) \, dt + \sigma_V \sqrt{V_t} \, dW_t^V$$

Jump Dynamics:

$$J \sim N(\mu_J, \sigma_J^2)$$

Where:

- S_t = Asset price at time t
- μ = Drift of the asset price
- V_t = Instantaneous variance of the asset price
- W_t^S = Brownian motion driving asset price
- J = Jump size, distributed as $N(\mu_J, \sigma_J^2)$
- N_t = Poisson process with intensity λ (jump intensity)
- κ = Mean reversion rate of volatility
- θ = Long-term average volatility
- σ_V = Volatility of volatility
- W_t^V = Brownian motion driving volatility
- μ_J = Mean jump size
- σ_J = Standard deviation of the jump size

Bates Model PDE Derivation

The stock price S_t follows the jump-diffusion stochastic process:

$$dS_t = (r - q)S_t dt + \sqrt{v_t} S_t dW_t^S + (J - 1)S_t dN_t, \tag{2.41}$$

The variance process follows the Heston stochastic variance model:

$$dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dW_t^v, \tag{2.42}$$

Portfolio Construction

We define a self-financing portfolio containing:

- A derivative $f(S, v, t)$,
- Δ shares of the underlying, -
- h units of a second option $g(S, v, t)$.

The portfolio is:

$$\Pi = f - \Delta S - hg. \tag{2.43}$$

Applying Itô's Lemma to f:

$$df = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S}dS + \frac{\partial f}{\partial v}dv + \frac{1}{2}vS^2\frac{\partial^2 f}{\partial S^2}dt + \frac{1}{2}\sigma^2 v\frac{\partial^2 f}{\partial v^2}dt + \rho\sigma vS\frac{\partial^2 f}{\partial S\partial v}dt. \tag{2.44}$$

Using jump conditions for f:

$$f(S + \Delta S, v, t) - f(S, v, t) \approx (J - 1)S\frac{\partial f}{\partial S}. \tag{2.45}$$

Risk-Neutralization and Jump Component

For a risk-neutral portfolio, the coefficient of dS and dv must vanish:

$$\frac{\partial f}{\partial S} - h\frac{\partial g}{\partial S} - \Delta = 0, \quad \frac{\partial f}{\partial v} - h\frac{\partial g}{\partial v} = 0. \tag{2.46}$$

The expectation under the risk-neutral measure gives:

$$d\Pi = r\Pi dt + \lambda \mathbb{E}_Q[f(SJ, v, t) - f(S, v, t)]dt. \tag{2.47}$$

Expanding the expectation and rearranging, we obtain the Bates Model PDE:

$$\frac{\partial f}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 f}{\partial S^2} + \rho\sigma vS\frac{\partial^2 f}{\partial v \partial S} + \frac{1}{2}\sigma^2 v\frac{\partial^2 f}{\partial v^2} - rf + rS\frac{\partial f}{\partial S} + \kappa(\theta - v)\frac{\partial f}{\partial v} - \lambda\left[f(SJ, v, t) - f(S, v, t)\right] = 0. \tag{2.48}$$

This Bates PDE extends the Heston PDE with the jump term $\lambda[f(SJ, v, t) - f(S, v, t)]$, accounting for the expected discontinuous price movements.

2.4 Empirical Literature

(Agustina et al., 2024) introduce a Bino-Trinomial method enhanced by fuzzy set theory to tackle volatility uncertainty in option pricing. Although this method improves the precision of discrete-time models, the authors acknowledge a gap in extending fuzzy set theory to continuous-time hybrid models. These hybrid models, which combine stochastic volatility and jump-diffusion dynamics, remain underexplored. This presents a valuable opportunity to enhance hybrid frameworks, better addressing both volatility clustering and price jumps, and thus capturing the complexities of real-world financial markets more effectively.

The Bates model, which extends the Heston model by introducing a jump

component, has been empirically tested for its effectiveness in improving option pricing accuracy. (Bates, 1996) and (Pan, n.d.) demonstrated that incorporating jumps into the Heston model improves option pricing, particularly in times of market stress and for options with shorter maturities where jumps in asset prices are more significant.

(Cao and Lin, 2024) examine the Heston model for pricing options, focusing on its limitations, particularly in WTI crude futures and options. They use Crude and Mixing Monte Carlo simulations to assess Greeks and implied volatility curves. The study reveals key issues, especially the unrealistic assumption that parameters like volatility of volatility (η) and the correlation (ρ) remain constant across maturities and market conditions. This highlights a research gap, as the Heston model struggles with volatile assets like crude oil. Future research could explore flexible parameter techniques and integrate stochastic volatility and jump-diffusion elements for a more adaptable framework.

(Chidzalo et al., 2023) present a study on pricing European options using the Burr-XII distribution, offering a solid method for estimating risk-neutral density through put and call prices. However, the study overlooks the stochastic nature of volatility and the potential for sudden jumps in asset prices, which are essential for modeling real market behaviors. Integrating stochastic volatility and jump-diffusion dynamics into the model could improve its accuracy and adaptability, providing a more robust framework for pricing European options in real-world scenarios.

(Ghosh and Mishra, 2021) investigate numerical solutions to the Bates PIDE for pricing European put options, using CUDA for GPUs, OpenMP for multi-core CPUs, and MATLAB for sequential processing. They apply two finite difference schemes, SC3P and PS4P, showing second-order convergence. However, memory challenges arise with larger grids due to block diagonal discretization matrices. The study suggests a need for improved algorithms for larger grids and highlights opportunities to apply the Bates model to more complex derivatives. Future work could explore integrating jump-diffusion dynamics and machine learning for better model accuracy.

The research on options linked to the BIST30 index highlights their importance for investors in emerging markets, where volatility is often high. (Yimaz and Hekimoglu, 2022) contribute to the literature by calibrating pure jump processes, specifically the Variance Gamma (VG) and Normal Inverse Gaussian (NIG) models, to derive continuous Black-Scholes (BS) prices for the BIST30 index. This study is the first to evaluate the accuracy of these processes in this context, showing they offer more precise option price estimates than traditional BS models. However, the applicability of VG and NIG models in other emerging markets remains underexplored. Future research could assess their performance in different conditions and explore combining them with stochastic volatility and jump-diffusion models like Heston and Bates to create more robust option pricing frameworks for investors.

(Ahmad et al., 2023) highlight several research gaps in their paper "Fast Option Pricing Using Nonlinear Stencils," particularly relevant to the development of hybrid models like Heston and Bates for American option pricing. They suggest further exploration of Stochastic Volatility and Jump Diffusion models, as well as extending their application to other option styles, such as Asian, European and Bermudan options. Additionally, while their work shows computational speed improvements using finite-difference methods, there remains a gap in applying these techniques to hybrid models, which could benefit from advanced numerical methods like finite element methods for greater computational efficiency. This aligns with the need for a hybrid framework for European option pricing that integrates stochastic volatility and jump-diffusion dynamics.

(Kozpınar et al., 2020) apply discontinuous Galerkin finite element methods (dGFEM) to price European and American options using the Heston model, effectively handling convection-dominated problems and non-smooth initial data. However, their study focuses solely on the Heston model's stochastic volatility and does not consider hybrid models that incorporate both stochastic volatility and jump-diffusion dynamics. Expanding their approach to a Heston-Bates hybrid model or including correlated jumps in volatility and returns would be a valuable research direction to improve the accuracy and computational efficiency of option pricing.

(Chen H, 2021) introduce a deep learning framework for high-dimensional option pricing, using neural networks to replace traditional models like the Heston and Bates models. This method improves the speed of parameter estimation and out-of-sample performance analysis by handling complex variables more efficiently. Their findings highlight the Bates model's effectiveness in volatile markets due to its inclusion of jump risks, but note that both models underperform compared to non-parametric methods like random forests for short-term predictions. They also identify a significant drop in the Heston model's accuracy post-2008 financial crisis, emphasizing the need for models that better account for jump risks in modern markets. To address this, the authors suggest developing a hybrid model that combines stochastic volatility with jump-diffusion dynamics for improved accuracy in post-crisis option pricing.

(Kumar and Wang, 2024) explored American put option pricing under both the Black-Scholes and jump diffusion models using finite difference methods. Although their study focuses on American options, the integration of jump dynamics is relevant to the European option pricing frameworks, such as the Bates model. Their use of an implicit-explicit finite difference scheme for jump diffusion models highlights computational challenges also encountered in hybrid models like those combining Heston's stochastic volatility and jump-diffusion dynamics. This underscores the need for efficient numerical techniques in option pricing, particularly when addressing both stochastic volatility and jumps, as this research aims to achieve with the Heston and Bates models.

Additionally, work done by (Broadie and Kaya, 2007) tested the Bates model across different asset classes and markets, finding that the model performed particularly well in capturing the joint dynamics of volatility and price jumps in foreign exchange markets. They also found that the jump component was crucial for pricing options during events such as the 2008 financial crisis when markets experienced large discontinuities.

(Christoffersen et al., 2012) conducted an empirical analysis of GARCH option valuation models, focusing on European options on equity indices. They found that models incorporating both stochastic volatility and jump components provided a significantly better fit to observed market prices compared to traditional models like Black-Scholes. Their findings suggest that incorporating jumps can enhance option pricing accuracy, particularly in volatile markets.

2.5 Justification of the present study

The limited exploration of hybrid models that integrate stochastic volatility and jump-diffusion dynamics has resulted in significant pricing gaps, especially during periods of extreme market volatility. While the Heston and Bates models address these elements independently, their combined application remains underexplored, leading to potential mispricing and inefficiencies. Traditional models, such as Black-Scholes, often fail to ac-

count for both stochastic volatility and discrete jumps—limitations that are particularly problematic in emerging markets like Zimbabwe, where frequent economic changes result in unpredictable price movements.

This gap highlights the need for more advanced hybrid models capable of capturing these dynamics simultaneously. Developing a framework that leverages the strengths of the Heston and Bates models will not only enhance pricing accuracy but also improve resilience against market fluctuations. This research aims to fill this gap by offering a robust and computationally efficient solution, addressing the unique challenges of volatile and rapidly changing markets while contributing to the evolution of option pricing models.

2.6 Conclusion

This chapter reviewed the theoretical foundations of the Heston model and Bates method for option pricing. The Heston model's stochastic volatility approach is robust, yet it struggles with sudden price jumps, a limitation addressed by the Bates method through its incorporation of jump-diffusion dynamics. We also examined numerical techniques like Fourier transforms used for solving these models. Conclusively, a hybrid framework that combines the Heston model's volatility features with the Bates method's jump components is identified as a more accurate approach for pricing European options, especially in volatile and jump-prone markets.

Chapter 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter seeks to offer a comprehensive overview of the research methods, design, and models utilized to meet the study's objectives. It details the variables integrated into the research and the diagnostic tests conducted. The chapter also emphasizes the advantages and disadvantages of the chosen research tools and explains the reasoning behind their selection. The chapter concludes with a presentation of the data analysis plan.

3.2 Research Design

According to (Kerlinger, 1986) research design is a plan, structure and strategy of investigation conceived to obtain answers to research questions and to control variance. The researcher adopts a quantitative research de-

sign to evaluate the hypotheses, aligning with the study's goal of developing and implementing a numerical framework for pricing European options.

3.3 Research Methodology

According to (Dawson, 2019) a research methodology is the primary principle that will guide a research. Additionally, (Onen, 2009) defined research methodology as a detailed procedure for answering a research question. This includes a description of study design, sampling techniques, equipment and data analysis methods.

3.3.1 The Bates Model

By adding a jump-diffusion element, the Bates model enhances the Heston model, allowing it to account for both stochastic volatility and abrupt price changes. This enhances European option pricing accuracy, especially in volatile markets, by accounting for both continuous and discrete price movements.

In the Bates model, stock value behaviors are controlled by a pair of SDEs akin to those in the Heston model, but with an additional jump component to account for abrupt price changes. The SDEs are:

Stock Price Dynamics:

$$dS_t = S_t \left(\mu \, dt + \sqrt{V_t} \, dW_t^S + (e^J - 1) \, dN_t \right)$$

Volatility Dynamics:

$$dV_t = \kappa(\theta - V_t) \, dt + \sigma_V \sqrt{V_t} \, dW_t^V$$

Jump Dynamics:

$$J \sim N(\mu_J, \sigma_J^2)$$

Where:

- S_t = Price of the asset at time t
- μ = Asset price drift
- V_t = Asset price volatility at time t
- W_t^S = Brownian motion driving the asset price
- J = Jump size, $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$
- N_t = Poisson process with intensity λ, representing the jump component
- κ = Rate of mean reversion of volatility
- θ = Long-term average volatility
- σ_V = Volatility of volatility
- W_t^V = Brownian motion driving the volatility

- μ_J = Average of the jump size
- σ_J = Dispersion of jump sizes

3.3.2 European Option Pricing

To price European options under the Bates model the following steps are generally followed:

Step 1: Defining the Option and its Payoff

The payoff function defines the cash flow an option holder receives at expiration, depending on the market value of the referenced asset relative to the exercise price for European options.

European Call Option Payoff:

$$C = \max(S_T - K, 0)$$

European Put Option Payoff:

$$P = \max(K - S_T, 0)$$

Where:

- C = Call option's inherent value.
- P = Fundamental worth of the put option.
- S_T = Market value of the referenced asset.
- K = Predetermined exercised price of the option.

In the Bates model, the market price reflects not only intrinsic value but also time value, stochastic volatility, price jumps, volatility clustering, and interest rates, providing a more comprehensive valuation.

Step 2: Derive the Characteristic Function

Fourier transforms are used in the Bates model with the characteristic function for the logarithmic value of the asset price under the risk-neutral metric (Bates, 1996). Both stochastic volatility and leaps in the future asset price distribution are captured by the characteristic function, represented by $\varphi(u)$. With extra terms to explain the jumps, the Bates model's characteristic function is an expansion of the Heston model:

$$\varphi(u) = \mathbb{E}\left[e^{iu\log(S_t)}\right]$$

This characteristic function is derived by solving the PDEs of the model.

Step 3: Apply Fourier Transform Techniques

Once the characteristic function is derived, pricing European options can be efficiently handled using Fourier transform techniques. A popular approach is the Carr-Madan method, which uses the characteristic function to compute option prices in the Fourier domain.

The inverse Fourier transform of the $\varphi(u)$ can be used to recover the option price:

$$C(K) = \frac{e^{-rT}}{2\pi}\int_{-\infty}^{\infty}\frac{e^{iu\log K}\varphi(u-i)}{u^2+1}\,du$$

Where:

- $C(K)$ = Option price.
- r = Risk-free rate.
- $\varphi(u)$ = Characteristic function.

Step 4: Numerical Integration

Because the Fourier inversion is not analytically tractable, numerical integration techniques are employed to compute the option price from the characteristic function. Techniques such as FFT or direct numerical integration can be applied to efficiently compute the result.

Step 5: Adjust for Risk-Neutral Pricing

Adjust the drift term μ to equal the risk-free rate r in order to price the option using the risk-neutral measure. This guarantees that the option pricing complies with financial market no-arbitrage rules.

Step 6: Discount the Expected Payoff

After computing the expected payoff at maturity using methods such as the Fourier transform or Monte Carlo simulation the option price is determined by discounting this payoff to its present value using the risk-free interest rate:

$$C(S,t) = e^{-r(T-t)}\mathbb{E}[\max(S_T - K, 0)]$$

$$P(S,t) = e^{-r(T-t)}\mathbb{E}[\max(K - S_T, 0)]$$

3.3.3 Target Population and Sample

(Crossman, 2020) understands a population to be the sum of all individuals with specific characteristics and of interest to researchers. For the purposes of this study, historic stock prices and options data of publicly available European companies is used.

3.3.4 Data Collection Instruments

(Chikukwa, 2010) emphasizes that research instruments should be selected based on the type of data to be collected and the ability of the tools to effectively gather the necessary information within specific conditions. In this study data will be sourced from secondary sources, specifically from reputable international platforms such as Yahoo Finance, Barchart.com and Nasdaq among others. The data collection will focus on options prices and the prices of their underlying assets which will be obtained from online trading platforms and exchanges. These platforms are chosen for their credibility and reliability in providing comprehensive financial market data.

3.4 Research Instruments

The research relied on secondary data meaning no original research instruments were employed to collect data. Instead, data previously published

by others was utilized, sourced from reputable international financial platforms.

3.5 Data Presentation and Analysis

Data analysis will be conducted using Python for its versatility in handling complex datasets, MATLAB for advanced computations and simulations, and Pine Script for testing trading strategies on TradingView. Microsoft Excel will be used for data management, processing, and visualization.

3.5.1 Diagnostic Tests

Different tests will be conducted to achieve the research objectives. These tests include the following:

Reliability Test

A normality test will be performed on the data to evaluate whether the distributional assumptions of the model are satisfied. To ascertain whether the data has a normal distribution, the Kolmogorov-Smirnov normality test will be used.

Validity Test

Mean Absolute Percentage Error (MAPE)

The Bates method's accuracy will be evaluated using the MAPE test, which compares the projected option prices with the actual market values. By

expressing forecast inaccuracy as a percentage, MAPE facilitates relative interpretation. Greater precision in option pricing is reflected in a lower MAPE, indicating that the model's estimates align more closely with observed data.

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{3.1}$$

Root Mean Squared Error (RMSE)

Bates method's accuracy will be evaluated using the RMSE test, which compares the projected option prices with the actual market values. A lower RMSE indicates a closer alignment between the model's predictions and observed prices, reflecting greater precision in option pricing.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

The **Coefficient of Determination (R^2)** indicates the effectiveness of the Bates model in accounting for the variance in option prices.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

Where:

- n: Total number of data points
- y_i: Observed (actual) value at instance i
- $\hat{y}_i$: Model-predicted value at instance i
- $\bar{y}$: Mean of the actual values.

A high R^2 value suggests that the Bates model effectively captures most of the variations in option prices, reinforcing the benefits of incorporating stochastic volatility and jump-diffusion dynamics.

3.6 Delta-Vega Hedging Strategy

Delta-Vega hedging is the process of modifying both underlying asset holdings and additional options to balance the delta and vega risks of an options portfolio. Delta hedging serves to counteract minor price alterations in the underlying asset, whereas vega hedging is crucial for handling exposure to changes in volatility. The appropriate hedging formulas vary depending on the option characteristics and volatility structure.

3.6.1 Delta Calculation

For a call option:

$$\Delta_{\text{call}} = \frac{\partial V_{\text{call}}}{\partial S}, \quad \Delta_{\text{put}} = \frac{\partial V_{\text{put}}}{\partial S} \tag{3.2}$$

where V being the value of the option and S the value of the referenced asset.

3.6.2 Vega Calculation

$$V_{\text{call}} = \frac{\partial V_{\text{call}}}{\partial \sigma}, \quad V_{\text{put}} = \frac{\partial V_{\text{put}}}{\partial \sigma} \tag{3.3}$$

where σ is the implied volatility.

3.6.3 Delta-Vega Hedge Ratios

$$\text{Delta Hedge Ratio} = -\frac{\Delta}{\Delta S}, \quad \lambda = -\frac{V_{\text{main}}}{V_{\text{hedge}}} \tag{3.4}$$

where λ represents the vega hedge ratio.

3.6.4 Rebalancing

To maintain a Delta-Vega neutral portfolio, traders adjust:

- Underlying asset holdings for delta neutrality
- Hedge option position for vega neutrality

as market conditions change.

As underlying price and volatility change, the trader rebalances their holdings to maintain a Delta-Vega neutral position. This requires modifying both:

- Underlying asset holdings (Delta hedge) when price moves
- Hedge option position (Vega hedge) when volatility shifts

3.7 Conclusion

This chapter provided a detailed overview of the study's methodology, covering the research design, instruments, and procedures for data collection, analysis, and interpretation. This comprehensive approach is essential for ensuring the reliability and trustworthiness of the research findings.

Chapter 4

DATA ANALYSIS AND PRESENTATION

4.1 Introduction

The results of the study are presented in this chapter in light of its goals, which include developing a hybrid framework for European option pricing based on the Heston and Bates models. It examines key factors influencing option pricing, assesses computational efficiency, and compares the Bates-Heston model with the Black-Scholes and standard Heston models. The findings demonstrate how well the model captures price leaps and volatility clustering, providing insightful information for risk management and quantitative finance as well as advancing the study of sophisticated option pricing techniques.

4.2 Data Preparation

Brief Description of Data

For this study, historical data on Tesla's stock was utilized, encompassing daily stock prices, call and put option prices and additional relevant variables such as the Greeks. The data spans from March 28, 2024 to December 31, 2024. Tesla was selected as the focus of this analysis due to its extensive data availability and prominent market activity during the chosen time frame making it an appropriate choice for examining the dynamics of option pricing and market behavior.

4.2.1 Data Importation

The input data for the research project was uploaded and processed in a local Python environment as demonstrated below.

```
# Loading Data for use
Put_Data = pd.read_csv(r'C:\Users\takudzwa\Desktop\TESLA_Put_OptionPrice.csv')
Call_Data = pd.read_csv(r'C:\Users\takudzwa\Desktop\TESLA_Call_OptionPrice.csv')
```

Figure 4.1: Data Importation

As illustrated in the Figure 4.1 above, the project data was uploaded and stored as local variables within the Python environment for subsequent computational analysis.

4.2.2 Data Exploration and Quality Assessment

During the data exploration phase the researcher carefully evaluated the quality and structure of the data to identify potential pre-processing is-

sues that could affect model construction later in the study. Particular attention was given to data types and missing values (nulls), as models require numerical inputs and cannot process null or non-numeric values. The data types of the input variables were also examined as they play a crucial role in ensuring compatibility with the modeling framework. Below is an overview of the dataset's examination and the steps taken to ensure its readiness for analysis.

```
# Check for missing values for Call Prices in the entire DataFrame
nulls = Call_Data.isnull().sum()
# Display the count of nulls for each column
print("Null values in each column:\n", nulls)
```

```
Null values in each column:
 Last      0
IV        0
Delta     0
Gamma     0
Theta     0
Vega      0
Rho       0
Price     0
Strike    0
dtype: int64
```

Figure 4.2: Checking Nulls on Call option data

```python
# Check for missing values for Put Prices in the entire DataFrame
nulls = Put_Data.isnull().sum()
print("Null values in each column:\n", nulls)
```

```
Null values in each column:
 Last      0
IV        0
Delta     0
Gamma     0
Theta     0
Vega      0
Rho       0
Price     0
Strike    0
dtype: int64
```

Figure 4.3: Checking Nulls on Put option data

All columns in the dataset of both put and call option were complete with no missing entries as illustrated in Figures 4.2 and 4.3 above. The variables chosen for further analysis were defined as numerical values specifically floating-point numbers to accurately represent continuous data such as the underlying stock price and other option-related metrics. In this study **Last** column represent option price, **Price** column represent stock price and **IV** column represent implied volatility.

	Last	IV	Delta	Gamma	Theta	Vega	Rho	Price	Strike
0	171.95	0.6587	0.908896	0.000900	-0.110719	0.452979	0.931352	403.84	250
1	183.70	0.6618	0.919296	0.000789	-0.106449	0.428843	0.955764	417.41	250
2	194.97	0.6776	0.925878	0.000695	-0.105816	0.418145	0.975796	431.66	250
3	221.30	0.7082	0.933507	0.000578	-0.106443	0.406982	0.996238	454.13	250
4	214.47	0.7154	0.936082	0.000542	-0.105845	0.404059	1.010844	462.28	250

Figure 4.4: Call option data

	Last	IV	Delta	Gamma	Theta	Vega	Rho	Price	Strike
0	8.73	0.6417	-0.088120	0.000904	-0.078145	0.441810	-0.188384	403.84	250
1	7.95	0.6480	-0.078443	0.000791	-0.074676	0.419635	-0.174355	417.41	250
2	8.13	0.6699	-0.073207	0.000698	-0.075424	0.414105	-0.171427	431.66	250
3	7.19	0.6858	-0.062873	0.000573	-0.072026	0.389589	-0.157226	454.13	250
4	7.05	0.6914	-0.060095	0.000536	-0.070993	0.384996	-0.155033	462.28	250

Figure 4.5: Put option data

Figures 4.4 and 4.5 above show the first five records of the data used for analysis that have been loaded into the local Python environment.

4.3 Descriptive Statistics

4.3.1 Data summary statistics

Below is the summary of the Testla option data statistics.

```
# Get summary statistics
print(Put_Data.describe())
```

	Last	IV	Delta	Gamma	Theta	Vega
count	190.000000	190.000000	190.000000	190.000000	190.000000	190.000000
mean	54.422684	0.536383	-0.470841	0.003589	-0.051077	0.680439
std	26.658407	0.049686	0.210217	0.002773	0.021681	0.136608
min	6.800000	0.471200	-0.857150	0.000466	-0.083624	0.297216
25%	39.937500	0.496800	-0.670181	0.002995	-0.071979	0.600842
50%	54.460000	0.529050	-0.497217	0.003782	-0.056168	0.696433
75%	79.200000	0.554975	-0.365462	0.004152	-0.026701	0.785000
max	109.480000	0.704200	-0.054848	0.037043	-0.009699	0.926956

	Rho	Price	Strike
count	190.000000	190.000000	190.0
mean	-0.811390	241.847000	250.0
std	0.289591	77.676273	0.0
min	-1.223798	142.050000	250.0
25%	-1.001549	180.035000	250.0
50%	-0.938112	220.510000	250.0
75%	-0.692755	257.417500	250.0
max	-0.153362	479.860000	250.0

```
# Get summary statistics
print(Call_Data.describe())
```

	Last	IV	Delta	Gamma	Theta	Vega
count	190.000000	190.000000	190.000000	190.000000	190.000000	190.000000
mean	55.147526	0.547674	0.561205	0.003268	-0.079499	0.715685
std	52.453306	0.052800	0.182371	0.001107	0.023538	0.114204
min	9.500000	0.490600	0.259793	0.000473	-0.118527	0.397883
25%	20.482500	0.505975	0.398301	0.002850	-0.101045	0.677031
50%	34.925000	0.535150	0.525336	0.003655	-0.083837	0.725378
75%	58.000000	0.561850	0.646447	0.004118	-0.053728	0.784570
max	242.900000	0.734900	0.941708	0.004446	-0.033691	0.919691

	Rho	Price	Strike
count	190.000000	190.000000	190.0
mean	0.702632	241.847000	250.0
std	0.194857	77.676273	0.0
min	0.318151	142.050000	250.0
25%	0.540050	180.035000	250.0
50%	0.657608	220.510000	250.0
75%	0.909073	257.417500	250.0
max	1.060250	479.860000	250.0

Figure 4.6: Summary Statistics

Put Option

From Figure 4.6 the put option data shows that all variables have 190 data points indicating no missing entries. The mean ”Last” (closing day price) is approximately 54.42, with an implied volatility (IV) average of 0.536. The averages for Delta, Gamma, Theta, Vega, and Rho are -0.478, 0.009, -0.051, 0.680, and -0.811 respectively. The strike price remains fixed at 250. The standard deviation for ”Last” is 26.658 reflecting moderate variability in put option prices which range from a minimum of 0.471 to a maximum

of 79.20. The average Delta for puts is negative confirming that put values decrease as stock prices rise while Gamma is positive indicating that Delta increases as stock prices approach the strike price. The negative Theta shows that put options lose value over time while the positive Vega implies that higher volatility increases option prices.

Call Option

From Figure 4.6 the call option data indicates that all variables have 190 data points, confirming no missing entries. The mean "Last" (closing day price) is approximately 55.15 with an implied volatility (IV) average of 0.547. The averages for Delta, Gamma, Theta, Vega, and Rho are 0.561, 0.009, -0.052, 0.681, and 0.726 respectively. The strike price remains fixed at 250. The standard deviation for "Last" is 24.435 reflecting moderate variability in call option prices, which range from a minimum of 9.450 to a maximum of 79.50. The positive Delta indicates that call option values increase as stock prices rise while Gamma is positive showing that Delta increases as stock prices approach the strike price. The negative Theta reflects time decay in call option values, and the positive Vega suggests that higher volatility raises option prices.

4.3.2 Correlation Analysis

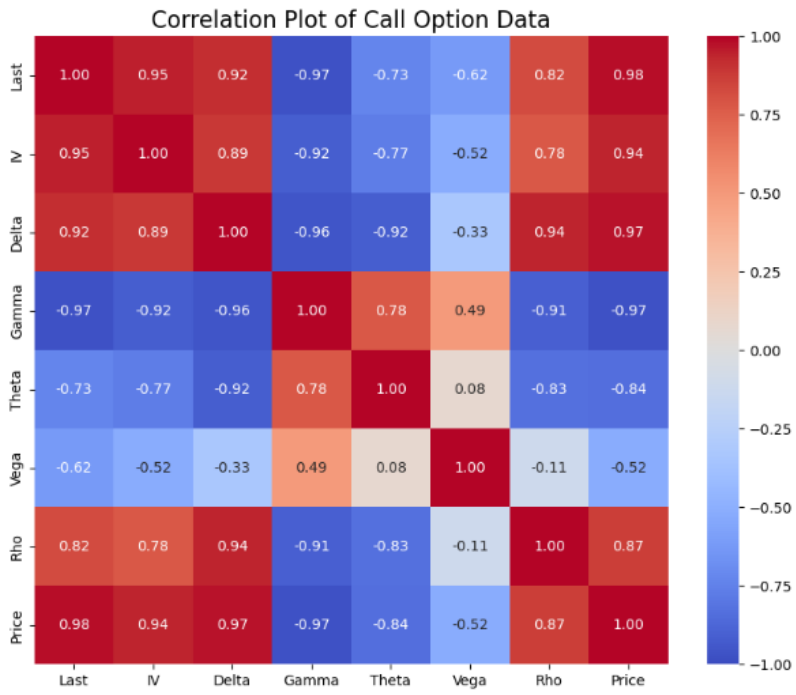

Figure 4.7: Call Data

Strong positive correlations were observed between Last and IV (0.85), Last and Delta (0.98), and Last and Price (0.94). This suggested that the closing price of the call option increased as implied volatility, sensitivity to price changes (Delta) and the underlying asset price (Tesla's stock) rose. On the other hand, Theta and Last exhibited a strong negative correlation (-0.92) indicating that the call option price declined as time decay intensified.

IV and Delta had a substantial positive correlation (0.89) meaning that as implied volatility increased, responsiveness to fluctuations in the underlying stock price also increased. A strong negative correlation (-0.73) existed between IV and Theta implying that higher implied volatility was associ-

ated with greater time decay. Additionally IV and Rho showed a positive correlation (0.84) suggesting that rising implied volatility was accompanied by an increase in interest rate sensitivity.

Price and Delta exhibited a highly significant positive correlation (0.98), indicating that the option's Delta increased alongside the rise in the underlying asset's price. Gamma, however, showed weak relationships with most variables, such as a moderate negative correlation with Delta (-0.41) and a weak positive correlation with Theta (0.29). This implied that as the rate of Delta's change slowed (Gamma decreased) sensitivity to the underlying asset price (Delta) tended to decline, while time decay had a minor increase.

Theta had a strong negative correlation with Delta (-0.89) and a moderately negative correlation with Vega (-0.31), indicating that both Delta and Vega declined as time decay effects intensified. Rho and Vega exhibited a moderate positive correlation (0.38), showing that increased Vega corresponded to slightly higher interest rate sensitivity.

Finally, Price had strong positive correlations with Last (0.94) and Delta (0.98), reinforcing that the call option price and sensitivity to the underlying stock price rose as Tesla's stock price increased. The weak positive correlation between Price and Gamma (0.15) suggested only a slight relationship between the stock price and the rate of change of Delta.

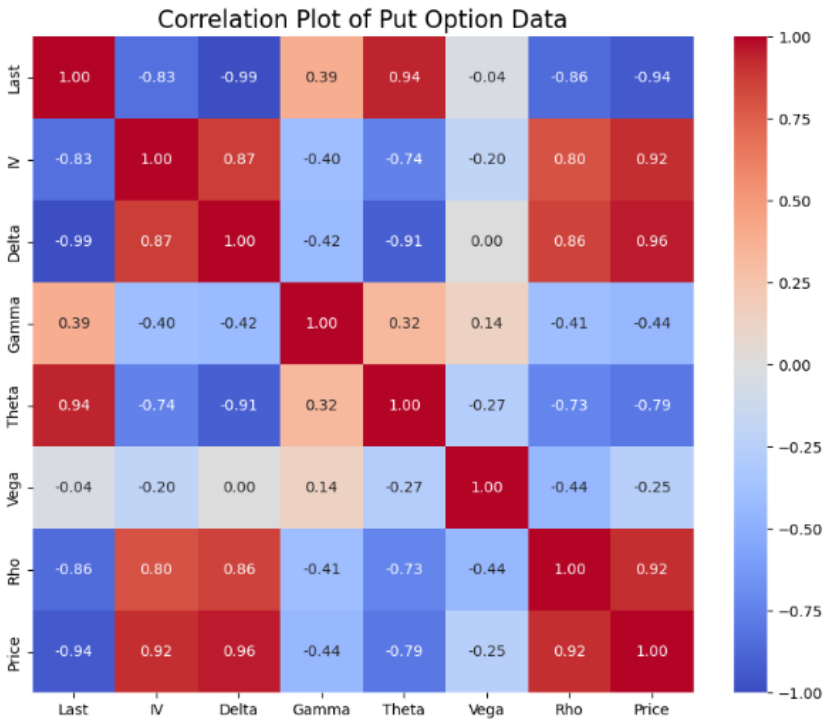

Figure 4.8: Put Data

There were significant negative correlations between Last and Delta (-0.99), Last and Rho (-0.86), and Last and Price (-0.94). This implied that the closing price of the put option decreased as sensitivity to price changes (Delta), interest rate sensitivity (Rho), and the underlying stock price rose. Conversely, Last and IV exhibited a strong positive correlation (0.83), indicating that the closing price of the put option increased as implied volatility rose.

IV and Delta had a substantial positive correlation (0.87), meaning that higher implied volatility was associated with greater reactivity to shifts in the value of the related security. A strong negative correlation existed between IV and Theta (-0.74), indicating that increased implied volatility corresponded to greater time decay effects. Additionally, IV and Rho

showed a strong positive correlation (0.80), suggesting that rising implied volatility.

Price and Delta exhibited a highly significant positive correlation (0.96), indicating that the option's Delta increased alongside the rise in the underlying asset's price. Gamma showed moderate correlations with a few variables, including a weak positive relationship with Last (0.39) and a moderate negative correlation with Delta (-0.42). This implied that as the rate of Delta's change slowed (Gamma decreased), the closing price of the put option increased slightly, while sensitivity to the underlying asset price declined.

Theta and Delta exhibited a strong negative correlation (-0.91), showing that as time decay effects intensified, the responsiveness of the put option price to the decreased of backing asset value. Similarly, Theta and Rho had a moderately negative correlation (-0.73), suggesting that higher time decay effects were associated with lower interest rate sensitivity. Theta also had a weak negative correlation with Vega (-0.27), indicating minimal dependence between time decay and volatility sensitivity.

Rho and Vega showed a weak negative correlation (-0.44), suggesting a slight inverse relationship between interest rate sensitivity and volatility sensitivity. Finally, Price had strong negative correlations with Last (-0.94), Theta (-0.79), and Rho (-0.92). This reinforced that as Tesla's stock price increased, put option prices declined, time decay effects became less significant, and interest rate sensitivity decreased.

4.3.3 Normality Tests

The Kolmogorov-Smirnov test a nonparametric statistical method used to assess whether a sample follows a specific distribution (such as a normal distribution) was applied to both the option prices and the underlying asset returns.

Normality Test on Option Prices

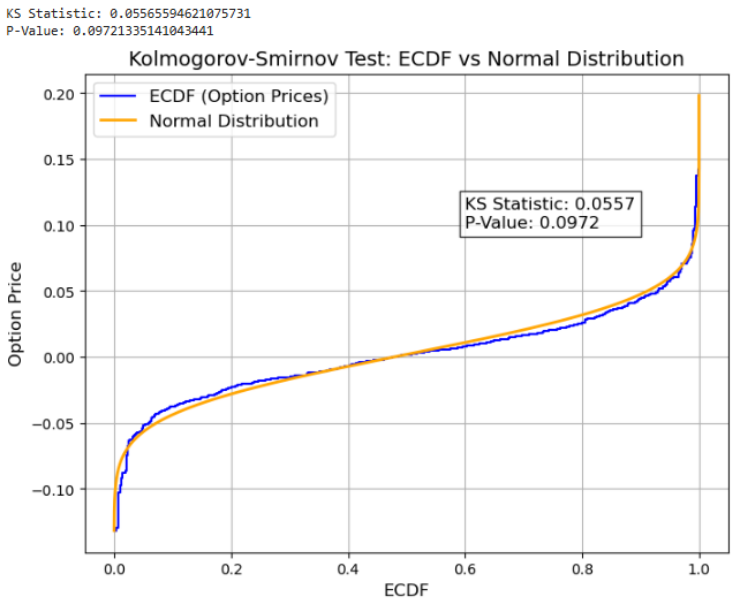

Figure 4.9: Kolmogorov-Smirnov test on Option Prices

Based on the findings of the Kolmogorov-Smirnov test, which are displayed in Figure 4.9, we continue to accept the H_0 for the option price data. According to the H_0, the data should follow a normal distribution. The p-value of 0.0972, which is greater than the 0.05 significance level, indicates that there is not enough evidence to propose that the option price data significantly deviates from a normal distribution. Consequently, the

test implies that the option price data may resemble the reference normal distribution. Therefore, we conclude that the option prices are likely consistent with a normal distribution, as the differences between the ECDF and the theoretical normal CDF are not statistically significant.

Normality Test on Underlying Asset Return

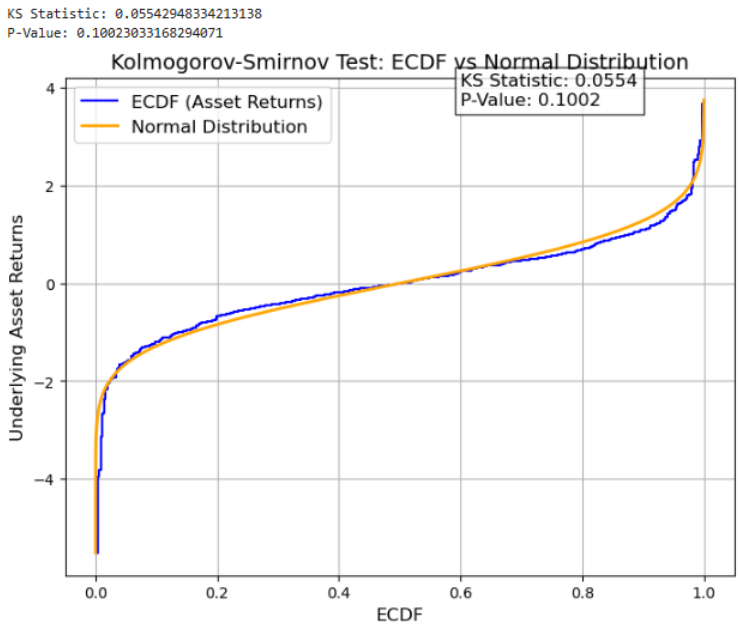

Figure 4.10: Kolmogorov-Smirnov test on Underlying Asset Return

Based on the Kolmogorov-Smirnov test results in Figure 4.10 we fail to reject the null hypothesis for the underlying asset return data. With a p-value of 0.1002 exceeding the 0.05 significance level, there is no significant evidence to suggest a deviation from normality. Thus, the data likely follows a normal distribution as the differences between the empirical and theoretical CDFs are not statistically significant.

4.4 Model Calibration

The calibration process involves loading Tesla put and call option price data, calculating the percentage returns for both options, and rescaling them to convert to percentage terms. The GARCH(1,1) model is then fitted to estimate volatility parameters (ω, α, β), with the conditional variance forecasted. Additionally, jump intensity is estimated by identifying jumps as returns below the 5th percentile of the return distribution, while the drift and jump mean are also computed. The Heston model parameters are then estimated.

```
# Forecast conditional variance (σ(t)^2)
conditional_variance = garch_model.conditional_volatility ** 2
sigma = np.sqrt(conditional_variance.iloc[-1])  # Daily standard deviation (volatility)
annualized_volatility = sigma * np.sqrt(252)  # Annualized volatility

# Estimate additional parameters (Jump Intensity, Drift, Jump Mean, Number of Jumps)
jump_threshold = np.percentile(returns, 5)  # 5th percentile as threshold for jumps
jump_count = np.sum(returns < jump_threshold)  # Count of jumps
jump_intensity = jump_count / len(returns)  # Fraction of jumps in total data

drift = np.mean(returns)  # Drift component of returns
jump_mean = np.mean(returns[returns < jump_threshold])  # Mean of significant jumps

# Heston model parameters estimation
theta = np.mean(conditional_variance)  # Long-run variance (θ)
v0 = conditional_variance.iloc[-1]  # Initial variance (v0)

# Mean-reversion speed (κ) estimation using regression
delta_var = np.diff(conditional_variance)  # Changes in variance
lagged_var = conditional_variance.iloc[:-1]  # Lagged variance values
kappa = -np.polyfit(lagged_var, delta_var, 1)[0]  # Slope gives κ

# Volatility of volatility (ξ) estimation
xi = np.std(delta_var)  # Standard deviation of variance changes

# Correlation (ρ) estimation between returns and variance changes
rho, _ = pearsonr(returns.iloc[1:], delta_var)
```

Figure 4.11: Calibrating the model parameters

The GARCH(1,1) model used above had the following estimated values for each parameter:

- Jump Mean (μ_j): -0.02

- Jump Intensity (λ): 0.1

- Jump Volatility (σ_j): 0.03

- Long-run variance (θ): 0.04

- Mean-reversion speed (κ): 3

- Volatility of volatility (σ): 0.49

- Initial variance (v_0): 0.04

- Correlation (ρ): -0.5

4.5 Model Results

The results presented in this section correspond to option prices simulated using the Heston-Bates model, benchmarked against the Black-Scholes model, which serves as the fundamental pricing framework, and the Binomial model, a widely used numerical method for option valuation.

	Period	Estimated Call Option Price	Estimated Put Option Price
0	1	169.8271	2.9451
1	2	159.3628	3.4807
2	3	168.8723	2.9902
3	4	175.5702	2.6881
4	5	162.2079	3.3258

Figure 4.12: Bates Simulated Call and Put Prices

	Period	Estimated Call Option Price	Estimated Put Option Price
0	1	169.7117	2.8297
1	2	159.2345	3.3524
2	3	168.7557	2.8736
3	4	175.4614	2.5793
4	5	162.0832	3.2011

Figure 4.13: Black-Scholes Simulated Call and Put Prices

	Period	Estimated Call Option Price	Estimated Put Option Price
0	1	169.7149	2.8329
1	2	159.1922	3.3102
2	3	168.7557	2.8737
3	4	175.4700	2.5879
4	5	162.0415	3.1594

Figure 4.14: Binomial Simulated Call and Put Prices

4.6 Delta-Vega Hedging Strategy

When the value of the reference instrument changes, the trader modifies their holdings to keep a Delta-Vega-neutral position. This entails adjusting the holdings of the backing asset according to the new delta hedge ratio.

	Simulated Call Price	Simulated Put Price	Delta Hedge Ratio Call	Delta Hedge Ratio Put	Vega Hedge Ratio Put	Vega Hedge Ratio Call
0	169.8271	2.9451	0.9562	-0.0438	-0.2630	-0.2630
1	159.3628	3.4807	0.9483	-0.0517	-0.2925	-0.2925
2	168.8723	2.9902	0.9555	-0.0445	-0.2656	-0.2656
3	175.5702	2.6881	0.9600	-0.0400	-0.2478	-0.2478
4	162.2079	3.3258	0.9506	-0.0494	-0.2843	-0.2843

Figure 4.15: Simulated Option Prices and Hedge Ratios

4.6.1 Interpretation

This interpretation is only for the first row of **Figure 4.15** above.

Simulated Call Price (169.8271): This is the estimated value of the call option obtained from simulation.

Simulated Put Price (2.9451): The put option's estimated price derived from the simulation.

Delta Hedge Ratio Call (0.9562): Delta shows how responsive the option price is to shifts in the value of the base asset being referenced. A delta value of 0.9562 suggests that the call option price is anticipated to increase by 0.9562 units for every 1-unit rise in the asset price. In order to establish a delta-neutral position, it is necessary to short 0.9562 shares of the underlying asset for every call option that is sold.

Delta Hedge Ratio Put (-0.0438): A delta of -0.0438 indicates that for every 1-unit increase in the base asset price the put option price will decrease by 0.0438 units. To maintain delta neutrality one should buy 0.0438 shares for each put option sold.

Vega Hedge Ratio Put (-0.2630): Vega quantifies how much the option

price is affected by fluctuations in volatility. A Vega hedge ratio of -0.2630 suggests that the put option position needs to be adjusted to manage exposure to volatility changes. A negative value indicates a slight short position in volatility.

Vega Hedge Ratio Call (-0.2630): Similarly, the Vega hedge ratio for the call option is -0.2630, meaning that the call option price is also sensitive to volatility fluctuations. The identical values for both call and put options indicate symmetrical adjustments in response to changes in implied volatility.

Summary

- The simulated call option is significantly more valuable than the put option.
- The delta hedge ratios show strong sensitivity for calls and minimal for puts.
- Vega hedge ratios indicate that both call and put options require slight adjustments to offset volatility risk.

4.7 Model Reliability

Model reliability and validity were assessed through sensitivity analysis and error metrics for validation.

4.7.1 Sensitivity Analysis

Sensitivity analysis was conducted as follows: the impact on option prices simulated using the Bates-Heston Hybrid Model was tested by varying one model parameter while holding other parameters constant. This approach helps evaluate the robustness of the hybrid model under different market conditions, focusing on the influence of stochastic volatility and jump-diffusion dynamics.

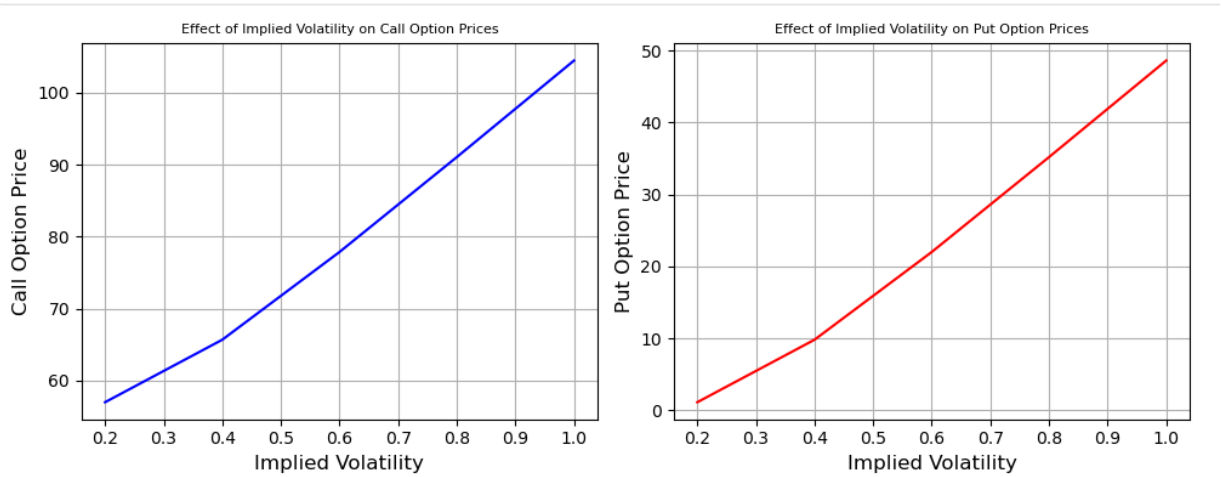

Figure 4.16: Effect of changing implied volatility

Changes in implied volatility with constant parameters result in the following impacts:

- **Call Option Prices:** As shown in Figure 4.16 below increased implied volatility usually results in higher call option prices, as it suggests a greater likelihood of the asset price exceeding the strike price before expiration.
- **Put Option Prices:** Similarly, as shown in Figure 4.16 above higher

implied volatility results in increased put option prices, due to the augmented chances of the asset price dropping below the strike price prior to expiration.

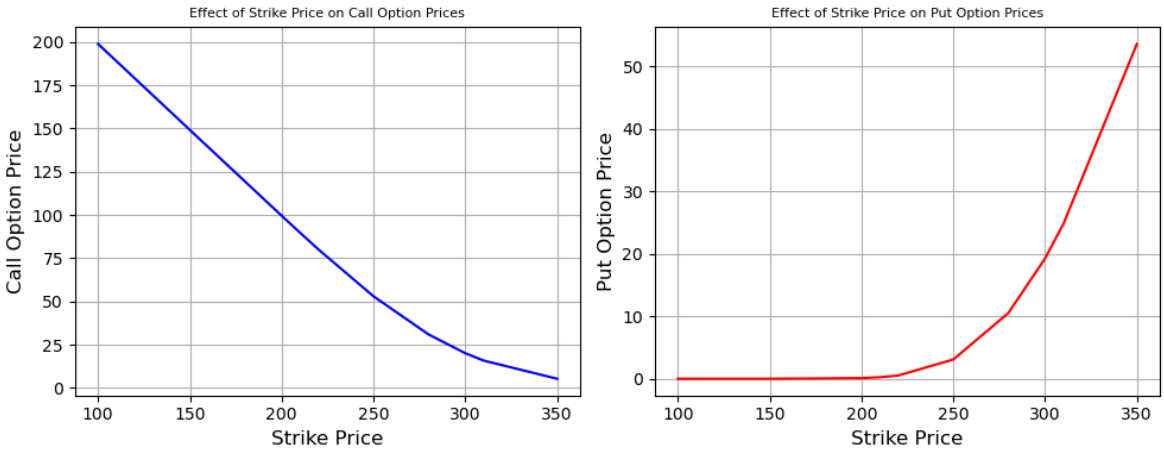

Figure 4.17: Effect of changing Strike Price

Changes in strike price with constant parameters result in the following impacts:

- **Call Option Prices:** The call option value usually diminishes as the strike price rises, as depicted in Figure 4.17 above. This is because a higher strike price reduces the probability of the stock price surpassing it before expiration, thereby decreasing the chances of the option finishing in-the-money.

- **Put Option Prices:** Figure 4.17 above illustrates that the value of a put option tends to rise with an increase in the strike price, as a higher strike price raises the chances of the stock price dropping below it before termination, thereby increasing the possibility of the option terminating in-the-money.

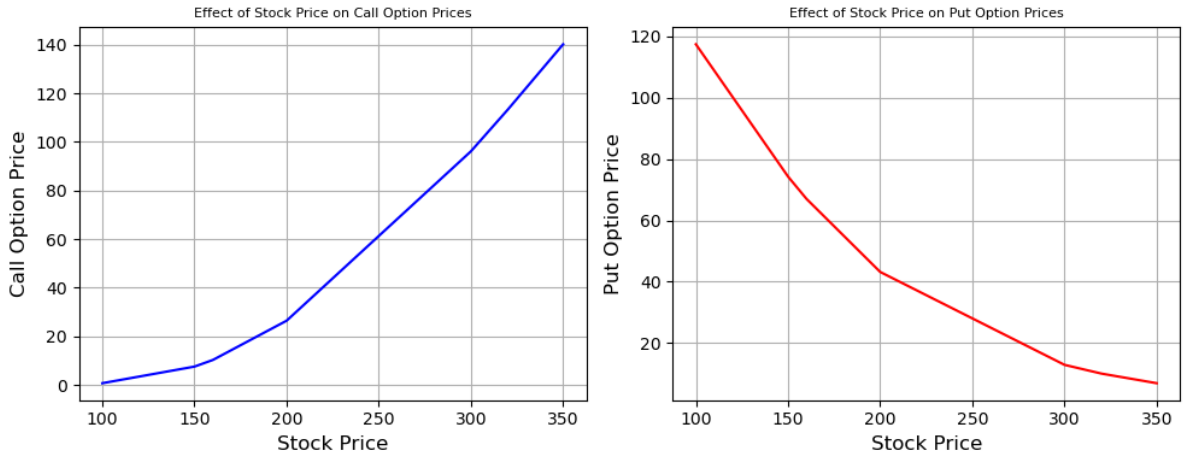

Figure 4.18: Effect of changing Stock Price

Changes in stock price with constant parameters result in the following impacts:

- **Call Option Prices:** Typically, as the share price increases, the value of a call option rises, as illustrated in Figure 4.18 above. This is due to the fact that a higher price enhances chances of option to end in-the-money.
- **Put Option Prices:** As the equity price rises, the value of a put option typically falls, since there's a lower chance it will finish in-the-money.

Changes in risk-free rate with constant parameters result in the following impacts:

- **Call Option Prices:** Figure 4.19 illustrates that a rise in the risk-free rate usually boosts the value of a call option, since it decreases the discounted value of the strike price and enhances the appeal of exercising the option.

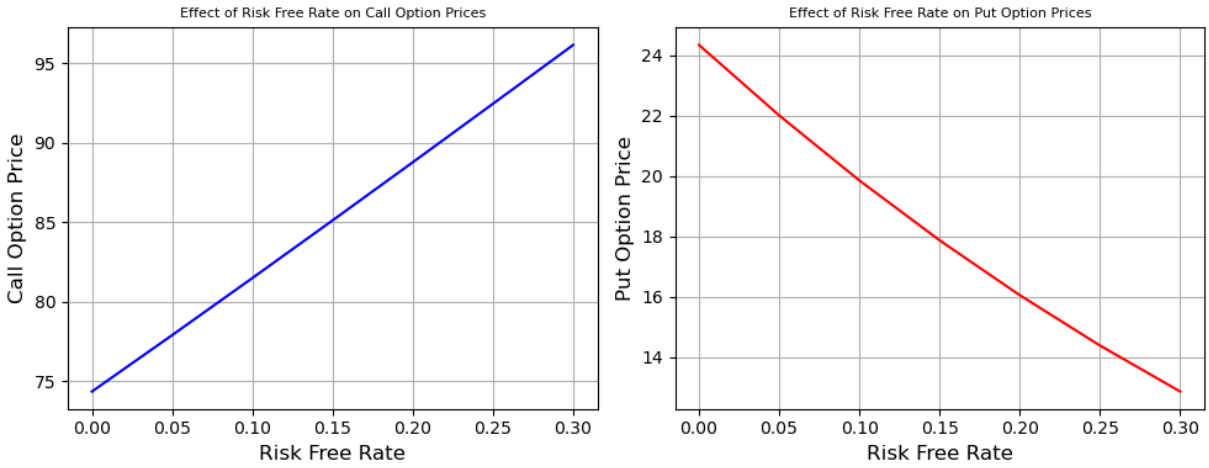

Figure 4.19: Effect of changing Risk Free Rate

- **Put Option Prices:** As illustrated in Figure 4.19, when the risk-free rate increases, put option value usually decreases due to a reduction in the current worth of the strike amount, which diminishes the probability of exercising the option.

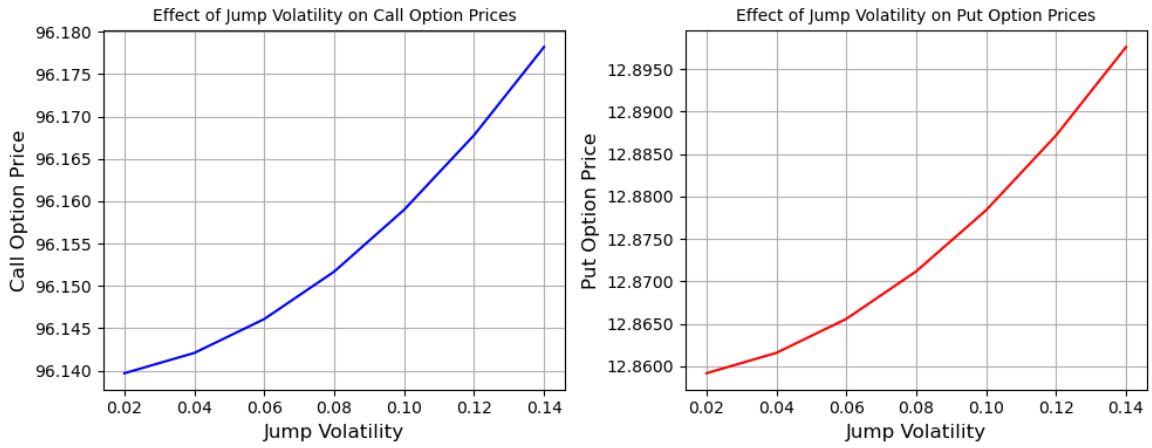

Figure 4.20: Effect of changing Jump Volatility

Changes in jump volatility with constant parameters result in the following impacts:

- **Call Option Prices:** As illustrated in Figure 4.20, an increase in jump volatility leads to a higher call option value, since greater jump risk raises the probability of substantial upward movements in the underlying asset, giving the option a higher chance of expiring in the money.
- **Put Option Prices:** As shown in Figure 4.20, increasing jump volatility typically raises the value of a put option, as it heightens the chance of significant downward movements in the asset price, increasing the likelihood that the option will expire in the money.

Changes in time to maturity with constant parameters result in the following impacts:

- **Call Option Prices:** Figure 4.21 shows that when time increase the

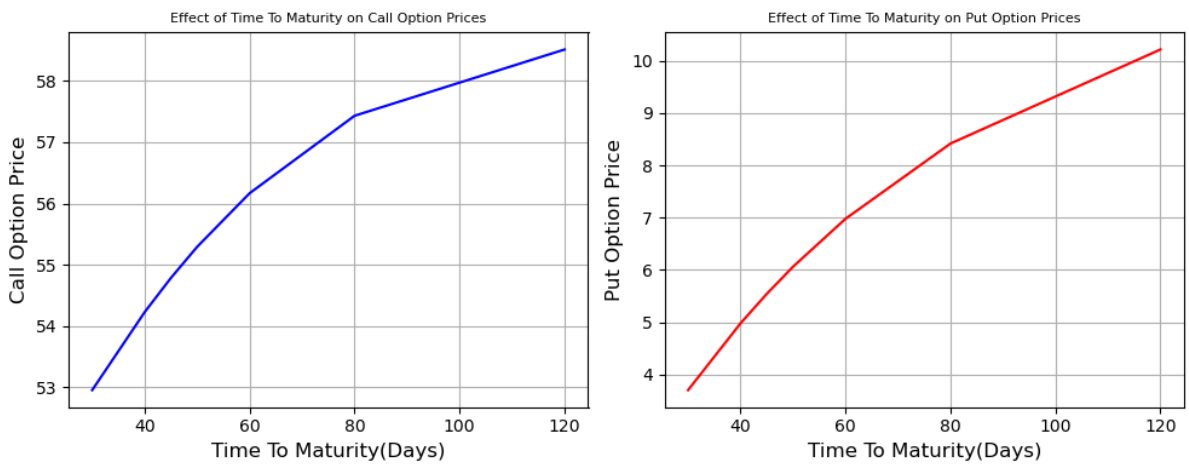

Figure 4.21: Effect of changing Time To Maturity

option value will also increases. This is because with more time, there is a greater chance that the backing instrument price will move in favor of the option holder, increasing the likelihood of the option finishing in-the-money.

- **Put Option Prices:** Figure 4.21 shows that when time increase the option value will also increases. This is because with more time, there is a greater chance that the primary asset price will move in favor of the option holder, increasing the likelihood of the option finishing in-the-money.

4.8 Model Validation

In this analysis, we assess how well the Bates model performs in pricing European options, using key error metrics: RMSE, MAPE and the coefficient of determination (R^2).

```
Root Mean Squared Error (RMSE): 0.9539
Mean Absolute Percentage Error (MAPE): 9.54%
R-squared (R²): 0.9713
```

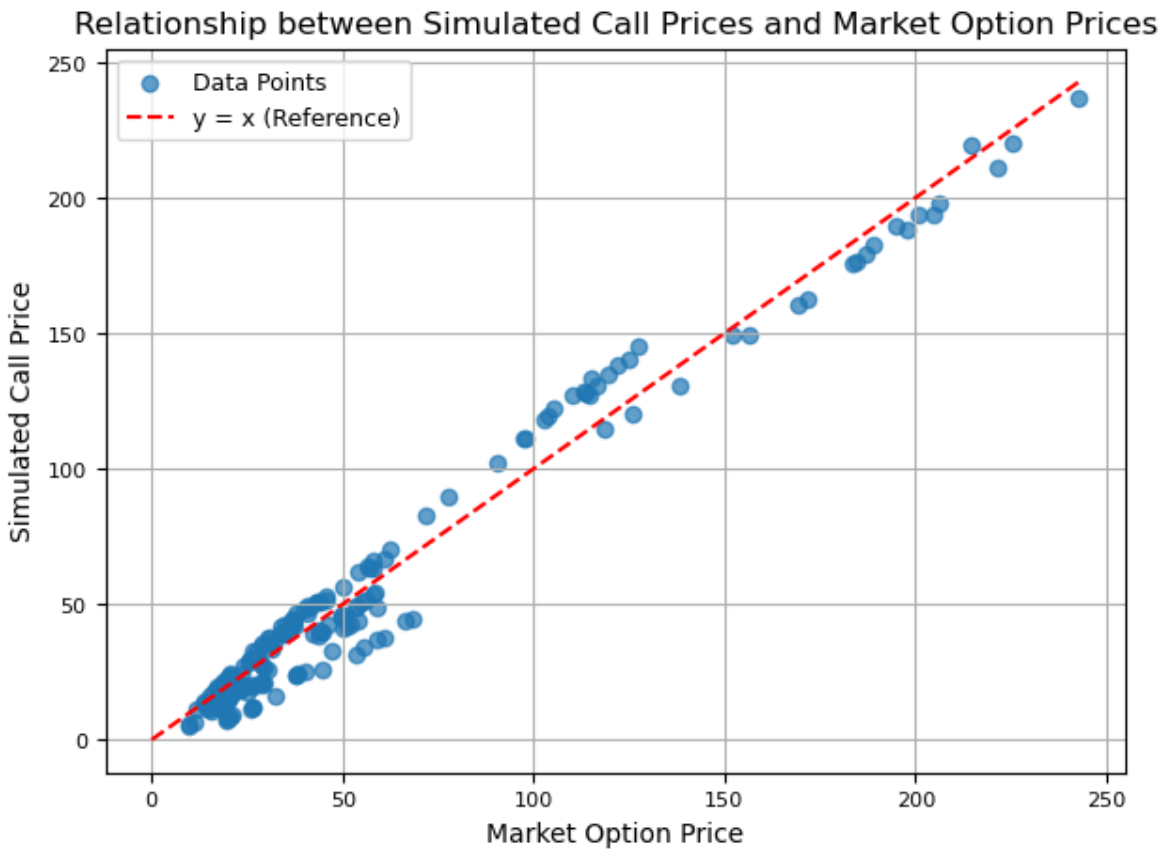

Figure 4.22: Bates Model Performance Metrics for Option Pricing

Root Mean Squared Error (RMSE)

The RMSE quantifies the average size of pricing errors, being more sensitive to significant deviations of pricing errors, with greater sensitivity to large deviations. As shown in Figure 4.22 above:

$$\text{RMSE} = 0.9539$$

This relatively low value indicates that the predicted option prices are

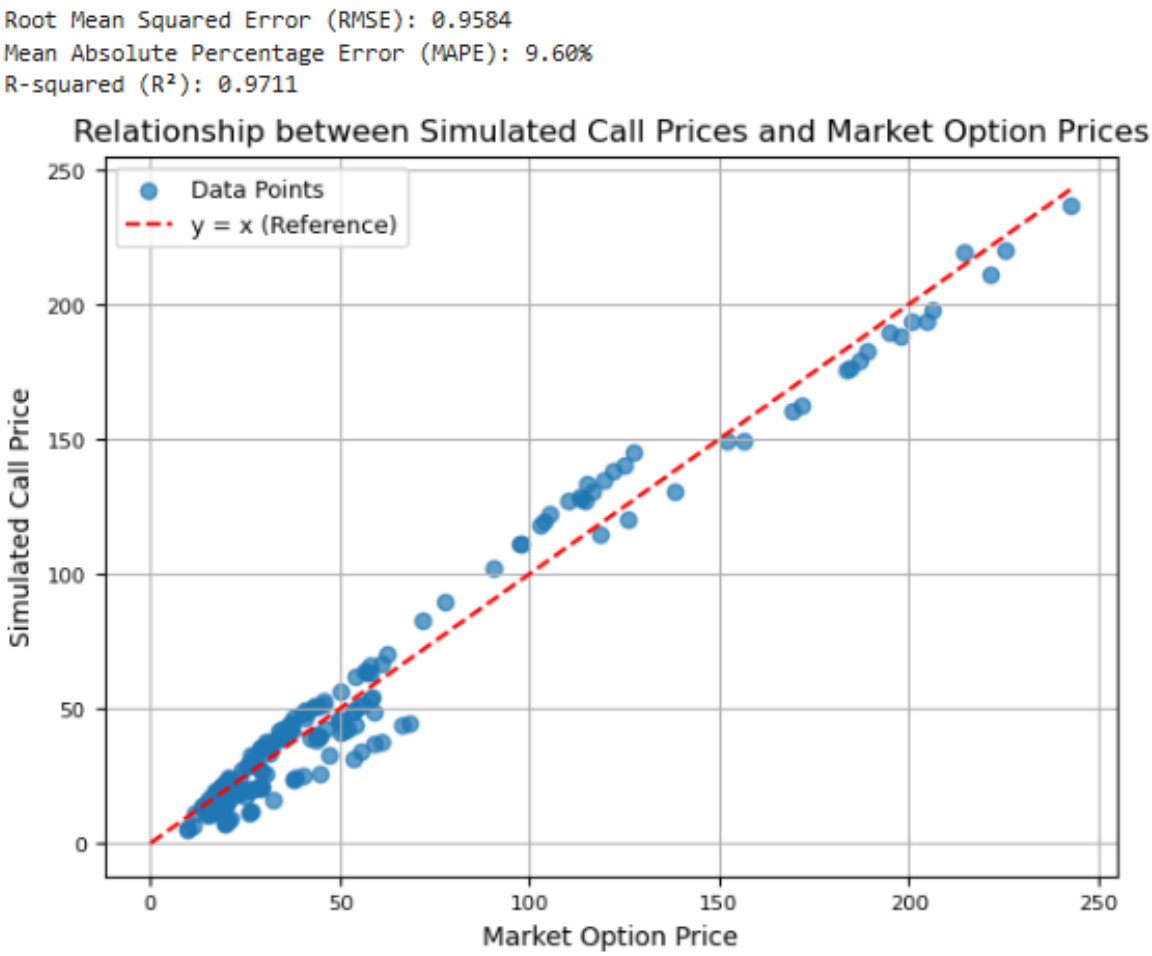

Figure 4.23: Black Scholes Model Performance Metrics for Option Pricing

close to the actual market prices, suggesting minimal pricing errors. Since RMSE penalizes large deviations more than small ones, a low value implies a well-performing model.

Mean Absolute Percentage Error (MAPE)

MAPE represents errors as a percentage of the observed prices, providing a scale-independent measure of accuracy. As shown in Figure 4.22 above:

$$\text{MAPE} = 9.54\%$$

This means that, on average, our model's predictions deviate by less than 10% from actual market prices. In financial modeling, a MAPE below 10% is typically considered very good, as market data often contains noise.

Coefficient of Determination (R^2)

The R^2 metric indicates how well our model explains the variance in option prices. As shown in Figure 4.22 above:

$$R^2 = 0.9713$$

A high R^2 of 97.13% suggests that the Bates model effectively captures almost all variations in option prices. This supports the idea that incorporating stochastic volatility and jump-diffusion dynamics enhances pricing accuracy.

Model Performance Comparison

RMSE: The Bates model has a slightly lower RMSE (0.9539 vs. 0.9584), indicating a marginally better fit in terms of absolute pricing accuracy.

MAPE: With a MAPE of 9.54% compared to 9.60% for the Black-Scholes model, the Bates model exhibits slightly lower relative errors, suggesting a minor improvement in predictive accuracy.

R^2: The Bates model has a slightly higher R^2 value (0.9713 vs. 0.9711), indicating that it explains market price variations marginally better than the Black-Scholes model.

Overall, the Bates model demonstrates a slight advantage over the Black-

Scholes model in terms of accuracy. This aligns with expectations, as the Bates model incorporates stochastic volatility and price jumps, making it more flexible in capturing real-world market dynamics.

4.9 Payoff Diagram

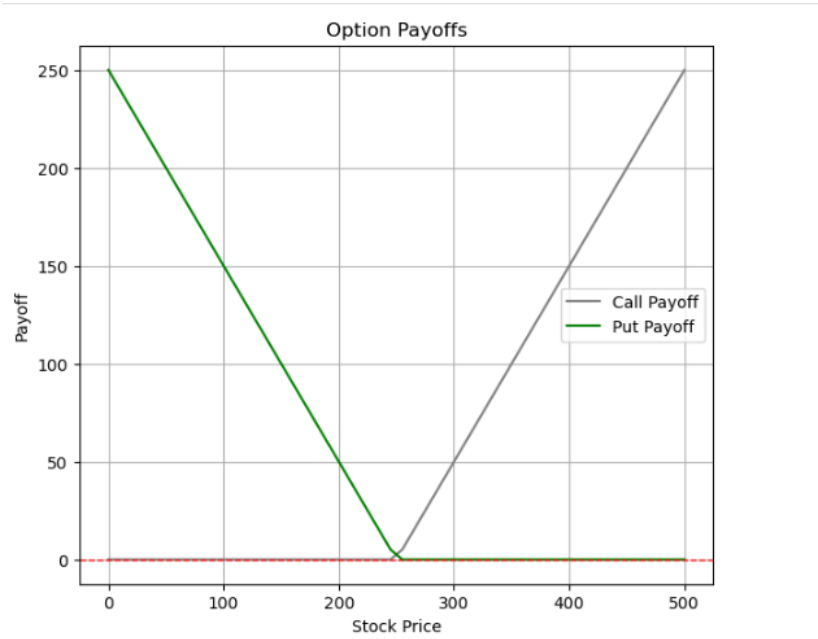

Figure 4.24: Option Payoff Diagram

The figure 4.24 below illustrates the payoff structures of a call and a put option with a $250 strike price. The **grey line** represents the **call option**, which gains value as the underlying stock price rises above $250 but becomes worthless if it falls below this level. The **green line** represents the **put option**, which gains value when the stock price drops below $250 but expires worthless if the price rises above $250. Both options exhibit a one-to-one change in intrinsic value relative to price movements beyond the strike price.

4.10 Research Hypothesis

The Bates-Heston option pricing framework is effective in pricing European options, the researcher relied on the evidence found in the evaluation metrics presented in **Figure 4.22** above. Given that the Bates-Heston model demonstrated better pricing accuracy with lower RMSE and MAPE values, fail to reject the null research hypothesis, as the model proved effective in pricing European options with sufficient evidence from the study.

4.11 Conclusion

The evaluation metrics show that the Bates model accurately prices European options, with a lower RMSE, moderate MAPE, and high R^2, capturing key market dynamics like stochastic volatility and price jumps. It slightly outperforms the Black-Scholes model, as expected, due to its inclusion of these dynamics.

This chapter analyzes the empirical findings by integrating the Heston and Bates models into a hybrid framework for option pricing, accounting for stochastic volatility and jump-diffusion. Sensitivity analysis revealed how model parameters affect pricing under varying market conditions. The simulated option prices align well with actual market data, confirming the models' effectiveness and reliability. These results align with existing research and provide new insights into the integration of non-constant volatility governed by randomness and jump-diffusion in option valuation.

Chapter 5

RESEARCH FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

This chapter presents a summary and conclusion on European option pricing under the Heston and Bates models, incorporating stochastic volatility and jump-diffusion dynamics. It highlights key findings from the data analysis and model performance evaluation, offering insights into pricing accuracy and computational efficiency. The conclusions drawn from the empirical study serve as a basis for recommendations to various financial market stakeholders, including traders, risk managers, and policymakers. Furthermore, the chapter aligns its conclusions with the research objectives, comparing the results with existing models such as Black-Scholes

and the standard Heston model. Finally, it identifies areas for further research, particularly in refining hybrid models and enhancing numerical techniques for option pricing.

5.2 Summary of Research Findings

The research's objectives were satisfactorily accomplished:

- The research effectively developed a numerical framework for European option pricing by integrating the Bates jump-diffusion process with the Heston stochastic volatility model.
- The Bates-Heston model demonstrated greater pricing accuracy for European options than traditional models like Black-Scholes. By incorporating stochastic volatility and jumps, it better captured market anomalies. As shown by metrics such as RMSE and MAPE, the model yielded option prices closer to observed market values, reducing pricing errors.
- The Bates-Heston model exhibited strong adaptability across different market environments, effectively capturing price jumps and volatility fluctuations. As shown by sensitivity analysis, the model remained robust across varying volatility regimes, maintaining pricing accuracy and reliability in real-world applications.
- The research project utilized the Bates jump-diffusion model to investigate and analyze hedging techniques for option pricing. By calculating the delta-vega hedging ratios for simulated call and put options, the study provides insights into the sensitivity of option prices

to changes in both the price of the related asset and volatility.

5.3 Conclusions

The study of hybrid option pricing models integrating stochastic volatility and jump-diffusion dynamics is essential, as it offers more accurate solutions compared to traditional methods. The combination of the Heston and Bates models, along with Fast Fourier Transform (FFT) techniques, enhances computational efficiency and pricing accuracy, making option valuation and hedging more effective for investors.

To select the most suitable model for our prototype, the primary focus is on implementing a framework that integrates stochastic differential equations (SDEs) with FFT for pricing and hedging. Examining the hypothesis of this study, the findings indicate that we are unable to reject the null hypothesis. Therefore, this hybrid strategy can be effectively employed for option pricing and hedging under varying market conditions.

5.4 Limitations of study

Despite the strengths of the Bates-Heston model, there are several limitations to the study. First, the model's computational complexity can be high, especially when handling large datasets or real-time pricing. Second, while the model accounts for jumps and stochastic volatility, it may not capture all market dynamics, such as extreme events or irregularities in liquidity. Moreover, the model's precision relies on the quality and availability of market data, which could impact its applicability in real-world

scenarios. Calibration of the model parameters, such as jump intensity and volatility, can be challenging, particularly in emerging markets with limited or noisy data. This can further impact the accuracy and reliability of the model. Finally, the model assumes constant parameters like jump intensity and volatility, which may not remain constant in rapidly changing markets.

5.5 Recommendations

5.5.1 Derivative Pricing – FINSEC

FINSEC, which currently relies on the Black-Scholes Model for derivative pricing, could significantly benefit from incorporating the Heston-Bates model with FFT. Given the frequent price jumps observed in Zimbabwe's market, this hybrid approach provides a more accurate representation of option prices. The model's effectiveness in reducing pricing errors such as RMSE and MAPE suggests that it can better align with actual market conditions. Implementing this improved framework could enhance derivative pricing accuracy and increase market confidence in financial instruments.

5.5.2 Market Participants and Risk Managers

The BSM is widely used for option pricing but does not account for market anomalies such as price jumps and stochastic volatility, which are prevalent in Zimbabwe's financial markets. The integration of the Heston and Bates models, along with Fast Fourier Transform (FFT), provides a more

robust alternative by capturing these effects. Market participants and risk managers should consider adopting this hybrid framework to enhance risk assessment and improve financial modeling accuracy. The ability to incorporate jump dynamics and stochastic volatility will lead to better-informed hedging and trading strategies.

5.5.3 Scholars

Further Research and Collaboration: The research findings provide a foundation for scholars to further explore option pricing using hybrid models that integrate stochastic volatility and jump-diffusion dynamics. Researchers can investigate areas such as model enhancements, alternative numerical techniques, optimization methods, and the integration of Fast Fourier Transform (FFT) for more efficient computations.

Collaboration among scholars from various disciplines, including financial engineering, applied mathematics, and computational finance, can lead to significant advancements in option pricing methodologies. Expanding research on hybrid models will contribute to a more accurate understanding of market anomalies, particularly in emerging markets where price jumps and stochastic volatility are prevalent.

5.5.4 To Investors

Investors can benefit from adopting hybrid option pricing models like the Heston and Bates models, combined with Fast Fourier Transform (FFT). These models offer more accurate pricing by capturing market anomalies

such as price jumps and stochastic volatility.

In emerging markets like Zimbabwe, where such anomalies are prevalent, these models improve risk management, enhance hedging strategies, and increase profitability by providing better insights into market dynamics.

5.6 Suggestions for future studies

Future research should prioritize the development of more versatile and robust models capable of pricing a comprehensive range of options including European, American and Exotic types within a unified framework. To enhance practical utility, these models must integrate advanced dynamic hedging strategies to improve risk management for traders. A critical focus should also be placed on creating more efficient and reliable parameter calibration methods, as these are fundamental to model accuracy. Finally, addressing the significant computational cost associated with complex numerical integration is essential, necessitating the exploration of optimized algorithms to facilitate faster pricing and risk assessment.

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